# Linear Voltage-Controlled Function Generator

**ELE504 – Major Design Project**

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# ABSTRACT

This project presents the design and implementation of the linear voltage-controlled function generator (VCFG) using discrete components (diodes, capacitors, resistors) and operational amplifiers. The purpose of this project was to produce a user-controlled square and triangular wave generator that has a variable frequency, linearly changing through a DC control Voltage ($V_c$) over a range of 0.1V - 5V, achieving two frequency ranges of 100 Hz - 3.5 kHz, and 20 Hz - 700 Hz. The overall system was brought up through four progressive milestones, starting with a fixed-frequency waveform generator, followed by a limiter and DC-to-±DC converter stages, and concluding with an integrated voltage-controlled oscillator with range and gain control. The designs and circuits were tested and validated through simulations in MultiSIM and breadboard testing that helped confirm the expected linear fo-Vc relationship, and amplitude controllability up to 8 $V_{pp}$. The overall design ended up meeting all major specifications with minimal discrepancy and demonstrated reliable frequency control, validating design approaches.

# OBJECTIVE & PURPOSE

The goal of this project is to design, model, implement, and test a Linear Voltage-Controlled Function Generator (VCFG) that can generate symmetrical triangular and square waveforms using just operational amplifiers and discrete components. The design must allow the user to change both frequency and amplitude and run only on ±12 V supplies, without any external input signal.

The primary objective is to create waveforms where the output frequency fluctuates linearly with the applied control voltage (Vc), without requiring any external signal input based on a proportionality constant K (Hz/V). Voltage-controlled frequency generation is crucial for modulation, data encoding, and signal synthesis applications in control, instrumentation, and measurement systems.

# INTRODUCTION

The initial design concept of the Linear Voltage Controlled Function Generator (VCFG) stems from the fundamental functionality of the operational amplifier and its many configurations. The challenge was to use these Op-amps to construct an analog circuit that would satisfy the technical specifications of producing square and triangular outputs while staying within a tight budget. The design requires the frequency of these waveforms to be linearly varying with an applied control voltage known as $V_{c,}$ while only operating on ±12V power supplies and other discrete components like resistors and capacitors.

The design approach heavily relies on two key Op-amp configurations, known famously as the comparator and integrator modules. The comparator module is based on a basic comparator, which was later developed into the Schmitt trigger version, which acts as the heart/square wave generator of the circuit. The initial signal stems from the comparator due to its regenerative properties, which produces a switching saturated signal that becomes the square wave of the circuit. That same pulse (square wave) is turned into a triangular wave through integration using the popular integrator module, alongside other submodules that are later mentioned and discussed in the report. Therefore, the basic design strategy is to integrate these two circuit blocks in a fashion that would enable the extraction of square and triangle waveforms, thereby enabling a voltage-controlled frequency.

# THEORY

### The Operational Amplifier

As mentioned earlier, the theory behind the Linear Voltage Controlled Function Generator stems from the fundamental principles of the operational amplifiers and its many useful configurations. The Op-amp IC takes the difference between its inverting and non-inverting inputs to output an amplified output of some magnitude A. This expression is normally recognized by.

$$Vo \ = \ A * (Vi^{+} \ - \ Vi^{+}) \ = \ A * Vd \qquad \text{(EQ.1)}$$

This equation forms the basis for all the Op-amp configurations used, such as the inverting bistable comparator and integrator circuits.

### The Comparator Module

To develop the inverting bistable comparator module, it is necessary to first understand the simple comparator as shown in **Figure 1.0**. The principle of operation of the comparator starts from the non-inverting input voltage. When the non-inverting input is greater than the applied voltage at the inverting terminal, the output of the comparator will ideally saturate at the positive power supply level (+Vcc Rail). The output will remain saturated until the non-inverting voltage goes below the inverting terminal's potential, at which then the output voltage will saturate at the negative rail (-Vcc). In **Figure 1.0,** the switching point/threshold is 0 volts, as the inverting input is set to ground. In other words, when the $Vi^{+}$ is greater than 0 volts, the comparator will saturate at +Vcc, and when below 0 volts, the output will be at -Vcc. This specific module is classified as a non-inverting comparator.

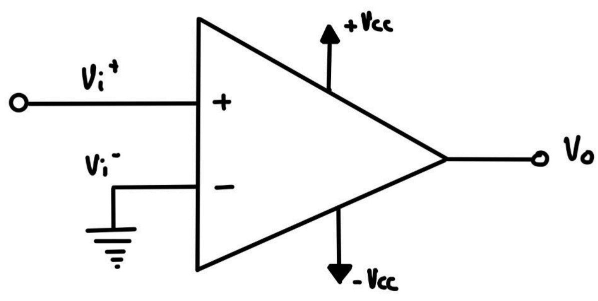


**Figure 1.0:** The Non-Inverting Comparator

Since the bistable comparator used is in its inverting form, it suggests that it stems from the inverting comparator of **Figure 1.1**. The inverting comparator acts very similarly to its non-inverting (**Figure 1.0**) counterpart, but the only difference is that its output is inverted. So when $Vi^{-}$ is greater than $Vi^{+}$, the output will saturate at -Vcc and vice versa.

In the specific case of **Figure 1.1,** the non-inverting terminal is grounded; therefore, the switching point will be 0 Volts.

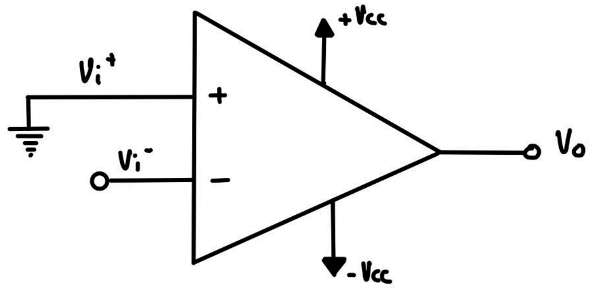


**Figure 1.1:** The Inverting Comparator

**The Inverting Bistable Comparator**

The inverting bistable comparator builds on the fundamental design of the inverting comparator. The only difference is that there is a positive feedback network instead of only grounding the non-inverting terminal.

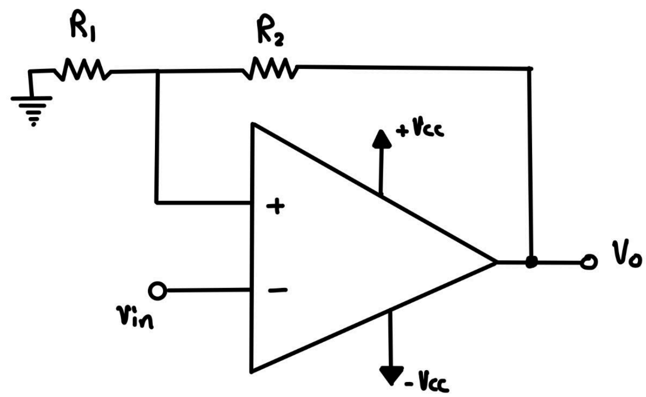


**Figure 1.2:** The Inverting Bistable Comparator

This positive feedback is used to create hysterysis. Unlike the basic inverting comparator that switches states at 1 specific threshold, the inverting bistable comparator defines two transition points to go from low to high and vice versa. This behavior enables noise immunity and ensures stable switching with low-frequency sources or noisy input signals. As shown in **Figure 1.2,** the bistable comparator has a voltage divider formed using $R_1$ and $R_{2,}$ which feeds back a portion of the output to the non-inverting input. This feedback creates a reference voltage that dynamically changes with the output potential. So when the output is at +Vcc, the reference at the non-inverting terminal is also positive, requiring the Vin to drop even lower to drive the output back to -Vcc and vice versa.

As a result, the threshold point is no longer 0 volts and can be derived at positive saturation using the voltage divider formula as:

$$V_{TH+} = \frac{R_1}{R_1 + R_2}(+\ Vcc) \qquad \text{(EQ.2)}$$

The same approximation can be done when the output is at the negative saturation stage, as:

$$V_{TH-} = \frac{R_1}{R_1 + R_2}(-\ Vcc) \qquad \text{(EQ.3)}$$

This way, the switching thresholds are established, and the voltage transfer characteristics can be graphed to better visualize the state-changing rhythm of the inverting bistable comparator (**Figure 1.3**).

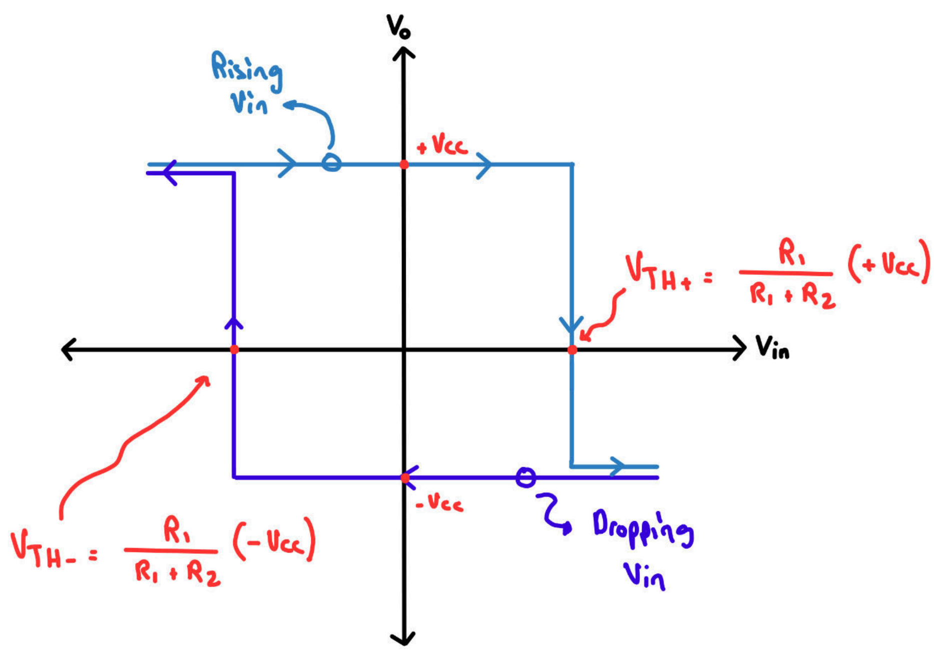


**Figure 1.3:** Voltage Transfer Characteristics (Inverting Bistable Comparator)

**When $V_{in} < V_{TH-}$:**

At first, the output is at the negative saturation level (-Vcc). As Vin decreases and crosses below the lower threshold $V_{TH-}$, the output instantly switches to the positive saturation (+Vcc).

**When $V_{in} > V_{TH+}$:**

At first, the output is at the positive saturation level (+Vcc). As Vin increases and crosses above the upper threshold $V_{TH+}$, the output instantly switches to the negative saturation (−Vcc).

Therefore, it is clear that the bistable comparator configuration is capable of generating a stable square waveform by switching between saturation levels. However, the square wave stage doesn't just end there, but requires an external driving event, such as the feedback signal from an integrator.

**The Integrator Module**

The next step in building the voltage-controlled function generator requires the integrator module. An integrator circuit is made using an Operational amplifier with a capacitor in its feedback path and a resistor at its inverting input. The figure below demonstrates the popular integrator circuit.

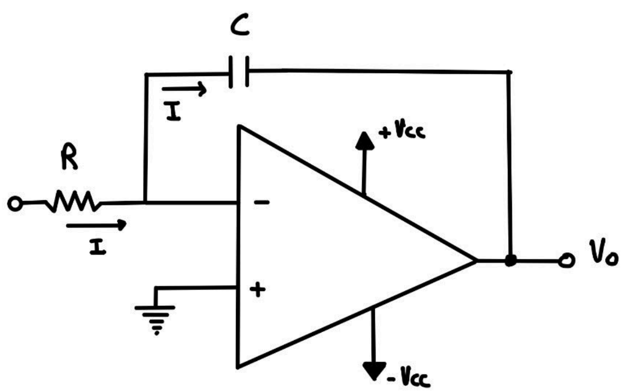


**Figure 1.4:** The Inverting Integrator

The integrator module is responsible for generating the triangular waveform of the circuit through the time integral of the input voltage fed into its inverting terminal. The operation of the module starts when a constant voltage is applied to the input, which drives current through the resistor (R), which charges or discharges the capacitor (C), causing the output Voltage ($V_o$) to ramp over time. **Figure 1.4** can be analyzed over a positive and a negative input.

**When Vin > 0:**

Using KCL at node $Vi^-$ results in

$$\frac{Vin - Vi^-}{R} = C * \frac{dVc}{dt} \qquad \text{(EQ.4)}$$

Observing the integrator module, it is visible that the inverting input of the op-amp is a virtual ground due to the negative feedback, resulting in $Vi^- = Vi^+ = 0V$. This simplifies Equation (4) to:

$$\frac{Vin}{R} = C * \frac{dVc}{dt} \qquad \text{(EQ.5)}$$

Vc is equivalent to Vc = 0 - Vo. Plugging that into Equation (5) and rearranging it results in:

$$V_o = -\int_0^t \frac{Vin}{RC} dt + Vc(0) \qquad \text{(EQ.6)}$$

Since the circuit is just powered on, that means the capacitor is fully discharged, so Vc(0) will be zero; therefore, the equation becomes:

$$V_o = -\frac{Vin}{RC} * t \qquad \text{(EQ.7)}$$

Since the integrator is inverting, it is relevant from Equation (7) that when the input voltage (Vin) is positive, the output voltage slope will be negative for that half. Now, when observing the circuit for the Vin less than 0, the only difference here is that Vin will be a negative value, which will cancel out the negative sign from Equation (7), so therefore:

**When Vin < 0:**

$$V_o = \frac{Vin}{RC} * t \qquad \text{(EQ.8)}$$

Take a look at **Figure 1.5** to get a visual understanding of how the integrator output would look based on a symmetrical square input, which will later be replaced by the bistable comparators' signal.

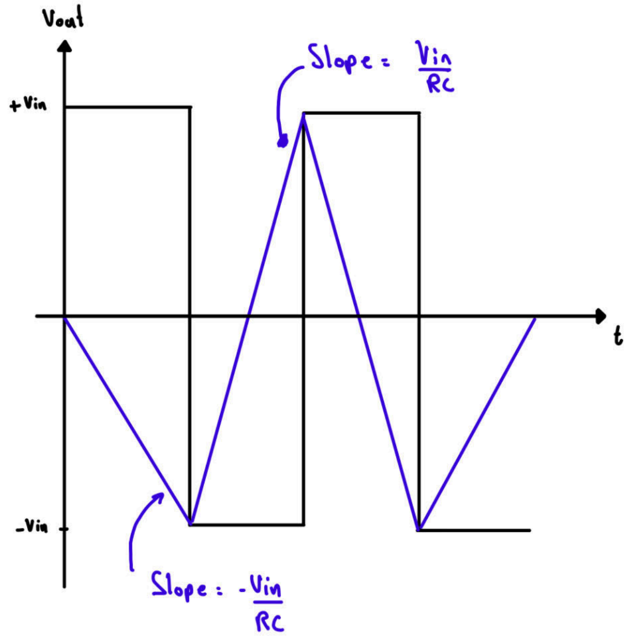


**Figure 1.5:** Integrator input and output waveforms

Looking at **Figure 1.5**, it begins to make sense that through the combination of the bistable comparators' square output signal and integrators' characteristics, it is possible to build a system capable of generating both triangular and square signals.

**Combining The Integrator & Bistable Comparator**

After verifying the core concepts of the integrator and the comparator modules, it is only now appropriate to proceed and analyze the combined circuits' characteristics of **Figure 1.6**.

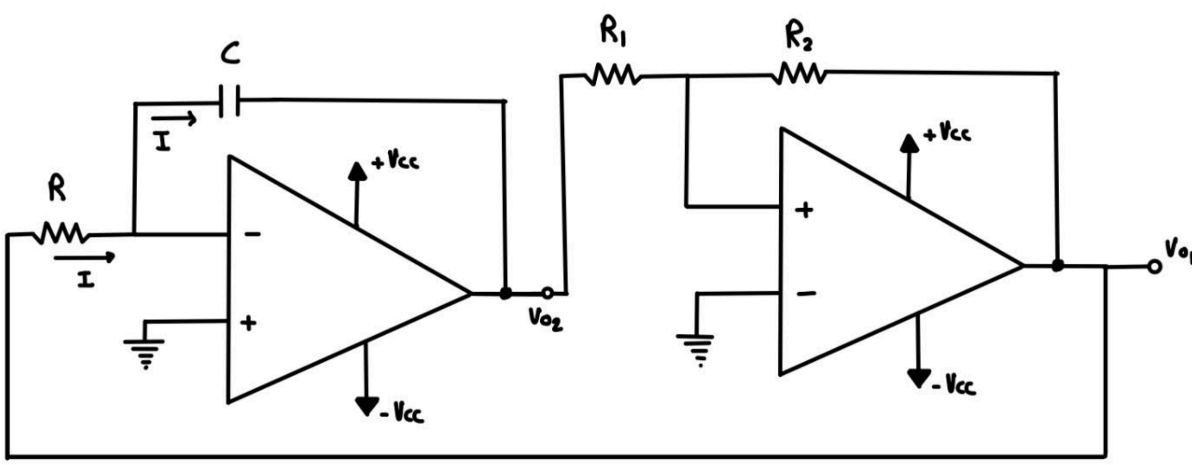


**Figure 1.6:** Combined Circuit

Notice how the comparator is in its non-inverting configuration. This is done because if not, the system would not oscillate. By making the bistable non-inverting, the loop creates positive feedback, which is essential for sustained switching between two states. This diagram shows the first step towards the final design. In this module, $V_{o1}$ indicates the triangular output with the amplitude of $V_{TH-}$ and $V_{TH+}$ as derived earlier. But this time, the input will be coming from the bistable comparator as input $V_{o1,}$ which is a square wave with the amplitude of +Vcc and -Vcc. The functionality of this circuit is pretty similar to the example of the inverting integrator with a function generator as input; the only difference here is that the bistable comparator will act as that input. Recall from **Figure 1.3** the inverting bistable comparator switches states when the input reaches particular threshold values known as $V_{TH-}$ and $V_{TH+}$. In the case of the non-inverting bistable comparator, the module switches to +Vcc when at $V_{TH+}$ and vice versa for the negative case. Now, to analyze the output $V_{o2,}$ suppose the bistable ($V_{o1}$) was at +Vcc, which means that the inverting input of the integrator is at +Vcc, suggesting that the output $V_{o2}$ will begin with a negative slope $-\frac{(+Vcc)}{RC}$. Notice how the output of the integrator is connected to the input of the bistable, which means that as that output keeps decreasing linearly, eventually it will drive the bistable to its negative threshold ($V_{TH-}$). In return, the bistable will produce/feed -Vcc potential, which will result in an integrator output ($V_{o2}$) of $-\frac{(-Vcc)}{RC}$ (Positive slope). The positive slope will continue until the integrator drives the bistable to its positive threshold ($V_{TH+}$), and the process will repeat. The figure below allows the visualization of this module's waveforms and assists in understanding the process.

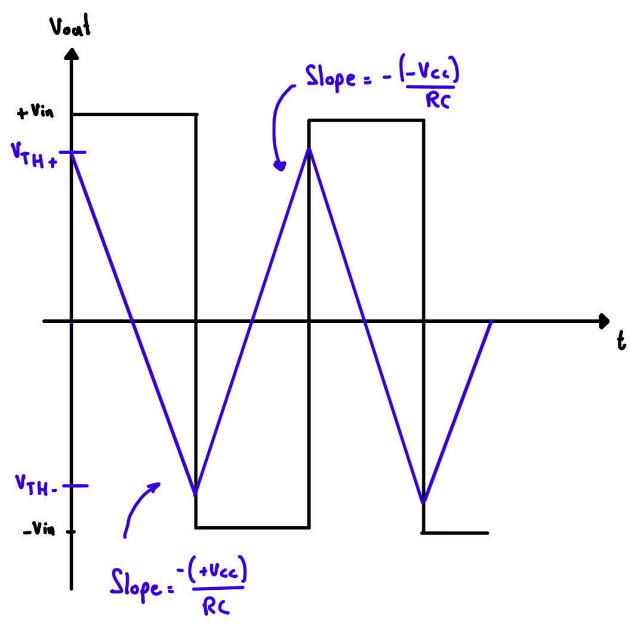


**Figure 1.7:** Integrator ($V_{o2}$) & Bistable ($V_{o1}$) Output waveforms

One of the fundamental requirements of the design is to have a linear relationship between the circuit frequency and control voltage (the control voltage has not yet been implemented). To achieve this linear behaviour, it is first necessary to derive the frequency equation of the circuit in **Figure 1.6** (combined circuit).

It is possible to derive the frequency expression by creating a relation between the period and the rising slope of the triangular wave. The red annotations in the figure below can help visualize the scenario.

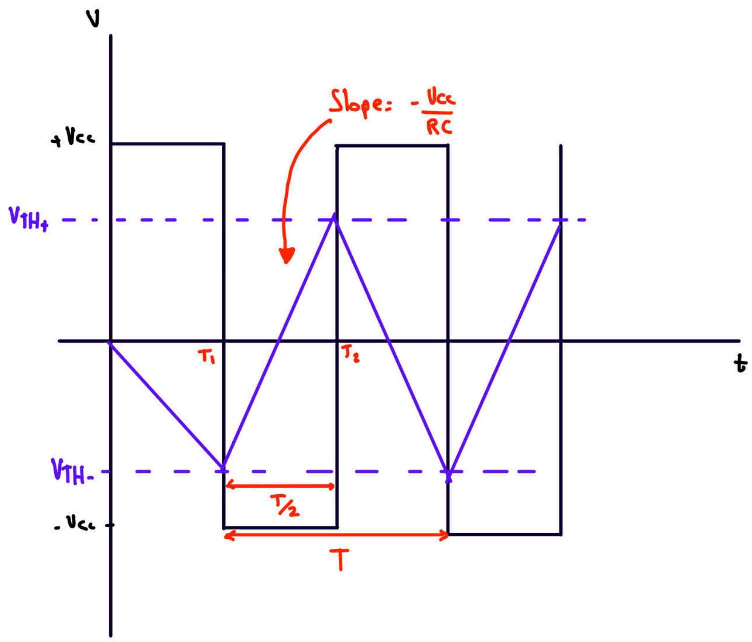


**Figure 1.8:** Combined Circuit Period Visualization Graph

The rising ramp occurs during the period between times T1 and T2 with the slope of -(-Vcc)/RC. To derive the slope equation using the period, it would be necessary to use the standard slope equation:

$$Slope \;=\; \frac{V_{TH+} - V_{TH-}}{T2 - T1} \qquad \text{(EQ.9)}$$

As derived earlier, it is also known that the slope of the positive ramp is expressed as

$$Slope \;=\; \frac{-(-Vcc)}{RC} \qquad \text{(EQ.10)}$$

Setting equations 9 and 10 equal to each other will result in:

$$Slope = \frac{V_{TH+} - V_{TH-}}{T/2} = \frac{-(-Vcc)}{RC} \qquad \text{(EQ.11)}$$

From equation(11), it only takes a bit of algebra to rearrange and extract the period expression:

$$T = \frac{2RC \times (V_{TH+} - V_{TH-})}{-(-Vcc)} \qquad \text{(EQ.12)}$$

Using the frequency equation (f = 1/T) to get the circuit's frequency expression:

$$f = \frac{1}{T} = \frac{-(-Vcc)}{2RC \times (V_{TH+} - V_{TH-})} \qquad \text{(EQ.13)}$$

**DC to ±DC Converter & Digital Switch**

To implement the input voltage control, the DC to ±DC converter is a great way to do this, as it allows the circuit to be in sync with the output state of the bistable comparator. The DC to ±DC converter functions on a digital switch using a BJT transistor, which turns on and off based on the bistable output.

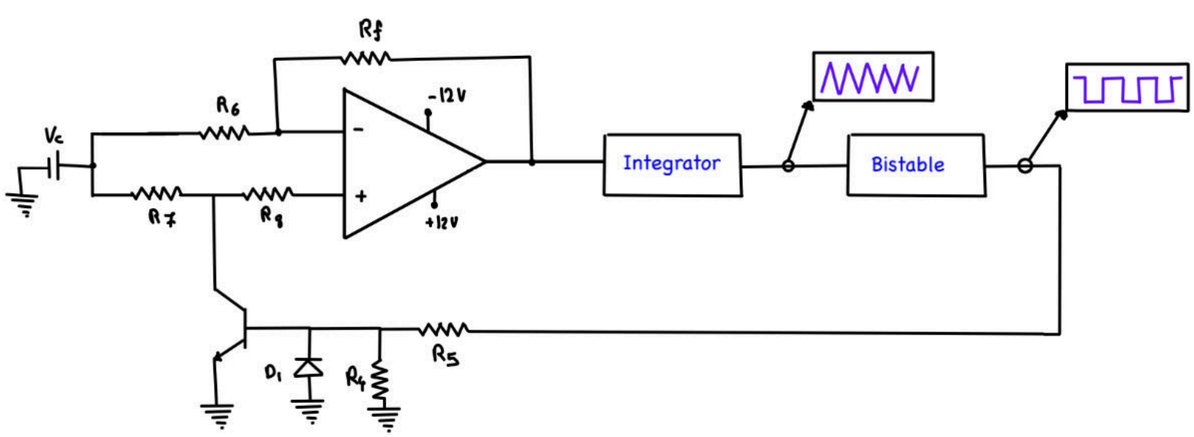


**Figure 1.9:** Combined Circuit With DC To ±DC

The analysis of the DC to ±DC converter can be separated into two parts, depending on the state of the switch.

**When BJT is ON:**

When the output of the bistable is at +Vcc, it drives the base of the transistor and forward biases the base-emitter junction, turning the BJT ON (saturation). In saturation mode, the collector-emitter path acts like a short to ground, which disables the resistor network around the Op-amp, causing the Op-amp to behave as an inverting amplifier. The output voltage expression can be given by:

$$Vo = -Vc$$

**When BJT is OFF:**

In this case, the Bistables output reverse-biases the BJT, causing it to turn off, and the diode (D1) clamps at -0.7 Volts to prevent the BJT from any damage. With the BJT on the feedback network no longer shorted, the op-amp behaves as a non-inverting amplifier, giving the output expression as:

$$Vo = Vc$$

Since all the resistors in the feedback path are intended to be equivalent, the gain expressions wouldn't really matter, as they would ultimately resolve down to an amplitude of 1, but for the sake of curiosity and understanding they can be derived as:

$$A_{Bjt-off} = (1 + \frac{R_F}{R_6})(\frac{R_8}{R_7 + R_8}) \qquad \text{(EQ.14)}$$

$$A_{Bjt-on} = -\frac{R_F}{R_6} \qquad \text{(EQ.15)}$$

With the implementation of the DC to ±DC converter, the circuit gained access to the user-controllable voltage, which now also influences the frequency expression of equation(13). Since the inverting input of the integrator is no longer fed from the bistable, the frequency equation(13) no longer relies on the rail voltages of the bistable's output, but relies on the DC to ±DC converter's output (Vc), therefore making up for the expression:

$$f = \frac{1}{2RC \times (V_{TH+} - V_{TH-})} * Vc \qquad \text{(EQ.16)}$$

This new frequency equation unlocks linearity between the control voltage and the frequency of the system, as when the control voltage is higher, the frequency of the circuit is higher as well, and vice versa. With this advancement, the first frequency range is made, and the frequency vs control voltage relationship can be visualized in the diagram below:

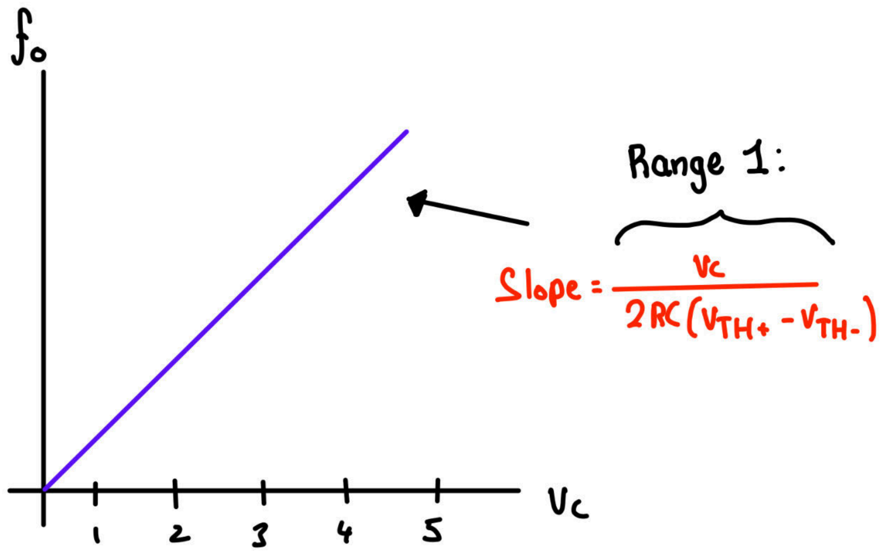


**Figure 1.10:** Frequency vs Control Voltage Relationship

**Voltage Limiting Diodes**

The voltage limiter diodes are small yet an important feature that assures greater stability of the system. The limiters are necessary to isolate the switch from the high voltages of the power supply and to simplify the design and maintainability of the overall circuit. The limiter is made out of two diodes in series, connected in opposite directions. The zener diodes limit the voltage at a constant rate (6.3V), which is more stable than the +Vcc/-Vcc supply rails of the bistables' output, as saturation values can fluctuate due to noise and other external factors. The limiter ensures the circuit always runs at one constant voltage as long as the bistables' output stays above the limiter's cutoff point, which is almost always the case.

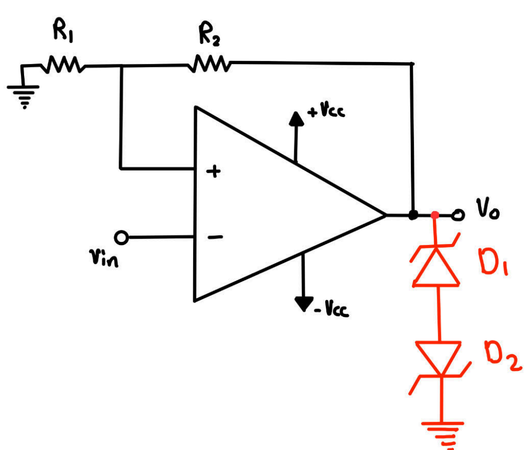


**Figure 1.11:** Bistable Comparator With Diode Limiters

The two cases for the Zener diodes operation are dependent on:

**When Bistable Output Positive:**

The upper diode (D1) is reverse-biased in zener breakdown, and the lower diode (D2) is forward-biased, resulting in:

$$V_{limiter+} = V_Z + V_f = 6.3V$$

**When Bistable Output Negative:**

The lower diode is reverse-biased in zener breakdown, and the upper diode is forward-biased, resulting in:

$$V_{limiter-} = -(V_Z + V_f) = -6.3V$$

**Gain Control & Multi-Ranged Frequencies**

The final two additions to the circuit unlock the desired 2nd frequency range and allow the user to control the output wave amplitudes with a voltage range. The two additions are a potentiometer in parallel with the switch (S1) that connects in series with the integrators' inverting inputs resistor. The second new addition to the circuit is the gain control implemented using two inverting gain amplifiers. The new addition can be shown in the color red in the figure below:

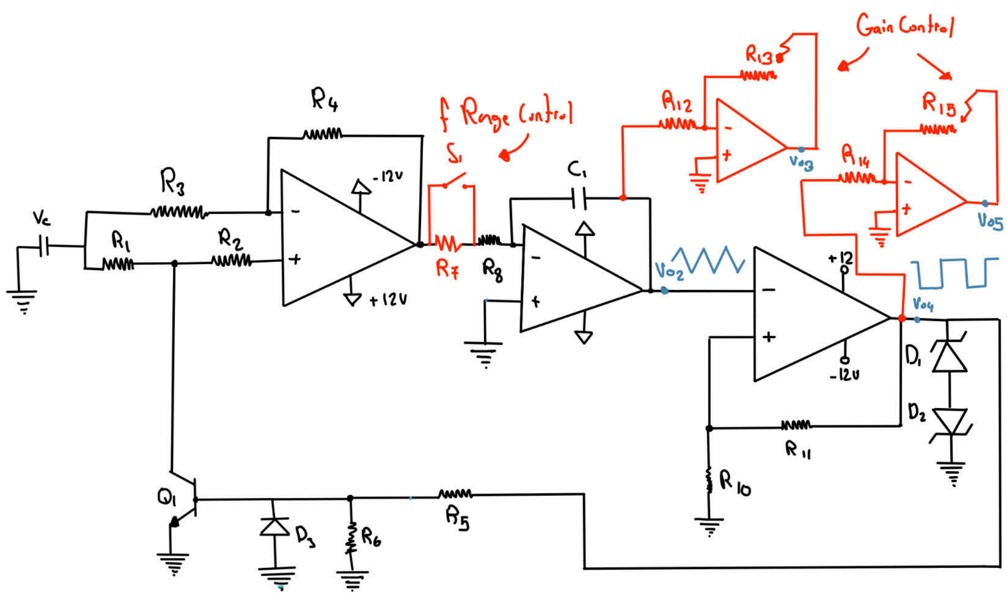


**Figure 1.12:** Final Design Of The Linear Voltage-Controlled Function Generator

These two final additions conclude the design of the linear voltage-controlled function generator, but they also make significant changes to some of the equations derived earlier. The first change is the 2nd frequency range that can be utilized now. The way to analyze this is through the state of the switch (S1):

**When Switch (S1) is closed:**

When the switch is closed, the resistor $R_7$ is shorted and acts like a wire, therefore setting the circuit in frequency range one as earlier derived from Equation(16). Equation 16 can be seen right below:

$$f_{Range\ \#1} = \frac{Vc}{2R_8C \times (V_{TH+} - V_{TH-})}$$

**When Switch (S1) is Open:**

When the switch is open, the resistor $R_7$ is no longer shorted and connects in series with the resistor $R_{8,}$ shifting the frequency to range #2.

$$f_{Range\ \#2} = \frac{Vc}{2(R_8+R_7)C \times (V_{TH+} - V_{TH-})} \text{ (EQ.17)}$$

The two frequency ranges can be displayed visually in the graph figure below:

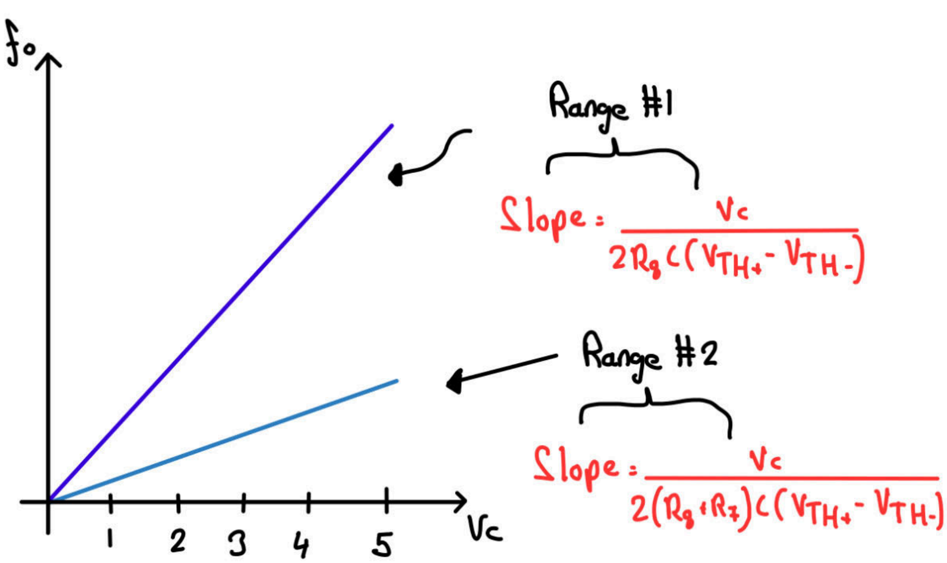


**Figure 1.13:** Frequency vs Control Voltage Relationship (Range 1 & 2)

The final new addition/gain stage can be analyzed easily using Operational amplifer circuit analysis techniques. As the gain stages are simply inverting gain amplifiers with negative feedback, there will be a virtual short, which can be analyzed by a KCL at the inverting nodes.

$$\frac{0 - V_{o2}}{R_{12}} + \frac{0 - V_{o3}}{R_{13}} = 0 \quad \& \quad \frac{0 - V_{o4}}{R_{14}} + \frac{0 - V_{o5}}{R_{15}} = 0 \qquad \text{(EQ.17)}$$

Rearranging these expressions gives us the gain expressions of:

$$V_{o3} = -\frac{R_{13}}{R_{12}} * V_{o2} \qquad V_{o5} = -\frac{R_{15}}{R_{14}} * V_{o3} \qquad \text{(EQ.18)}$$

With all of these derivations, the theory section of this report is concluded, and the operation behind the circuit is fully justified.

# Design Analysis

The overall design of the Linear Voltage-Controlled Function Generator integrates many important subsystems that all together contribute to the generation and control of the output waveforms. While in the theory section, it is established the underlying principles behind all of the modules (Comparator, Integrator, DC to ±DC Convert, etc), the design analysis section focuses on the actual design methodologies and the verification processes used to achieve the desired outcomes.

The Final design, as shown in **Figure 1.12,** is made up of 3 major functional blocks in a closed loop.

1. **Inverting Bistable Comparator:** Generating the square waveform by saturating between +Vcc and -Vcc.

2. **Integrator:** converts the square waveform into a triangular output through the continuous charging/discharging of the capacitor.

3. **DC to ±DC Converter:** creates the polarity of the control voltage (Vc), making the frequency dependent on Vc.

The 3 modules above are supported and regulated by the zener limiters to maintain circuit voltage stability, and include a range-select switch (S1) and a gain amplifier stage for user-adjustable frequency and amplitude. Combining these 3 major circuit blocks along with the support of the zener, gain control, and frequency range, we create a self-sustaining oscillator capable of generating triangular and square waveforms.

### Design Methodology

Each of the modules was designed individually before being combined into the full closed-loop system. The methodology of the design followed the incremental approach across 4 major milestones.

**Bitable Design:**

The schitt trigger/inverting bistable threshold was determined using the voltage divider formed by the resistors $R_{10}$ & $R_{11}$. These values were chosen to ensure the output of the comparator was at +Vcc & -Vcc, ensuring clean transitions and noise immunity. For this module, the LM318 Op-amp was used for its high slew rate and bandwidth, which are necessary when trying to accomplish sharp switching edges.

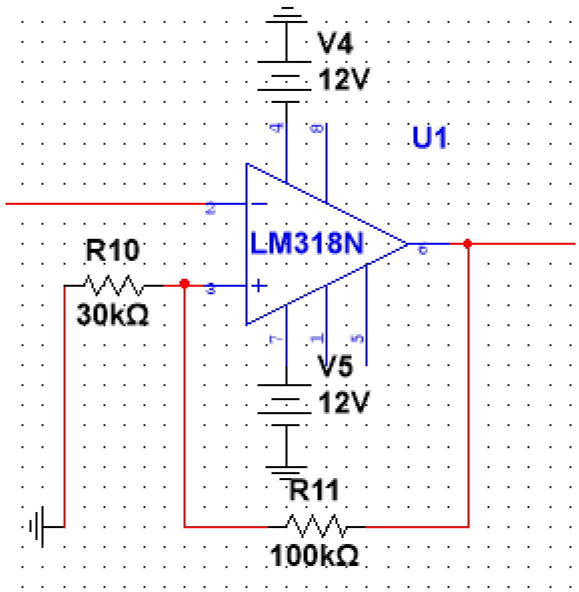


**Figure 2.0:** Bistable Comparator/Schmitt Trigger (MultiSIM)

**Integrator Design:**

The integrator's values were determined as $R_8$ = 25kOhms and C = 0.01uf, resulting in the desired frequency of 3.5 kHz at a constant voltage. The capacitor value was chosen based on component availability and ease of real implementation. The resistor value was then derived using the frequency equation by setting equation(13) equal to 3.5kHz and solving for the $R_8$ variable.

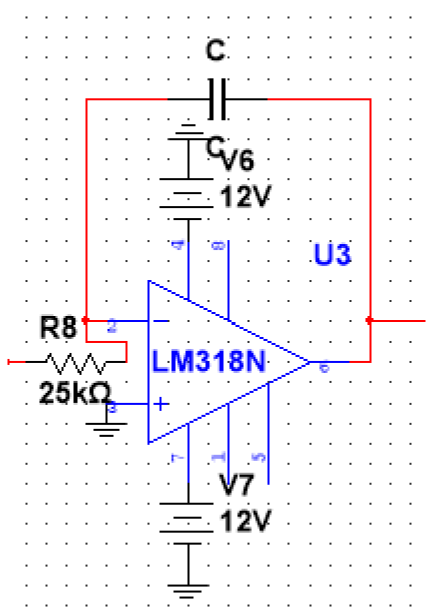


**Figure 2.2:** Integrators Design (MultiSIM)

**DC to ±DC Converter Design:**

This module was implemented using LM318 as well, and for the transistor, the 2N3904 model was used. When the bistable's output is positive, the transistor saturation, inverting the polarity of the control voltage (Vc), and when negative, it turns off the making it positive polarity. The design implements a diode at the switch to serve as a protection layer to avoid overdriving the transistor. The reason for this module is to introduce the user-controllable voltage (Vc). The values of all the resistors in this module were decided to be 1kOhm to avoid any gain in the output voltage.

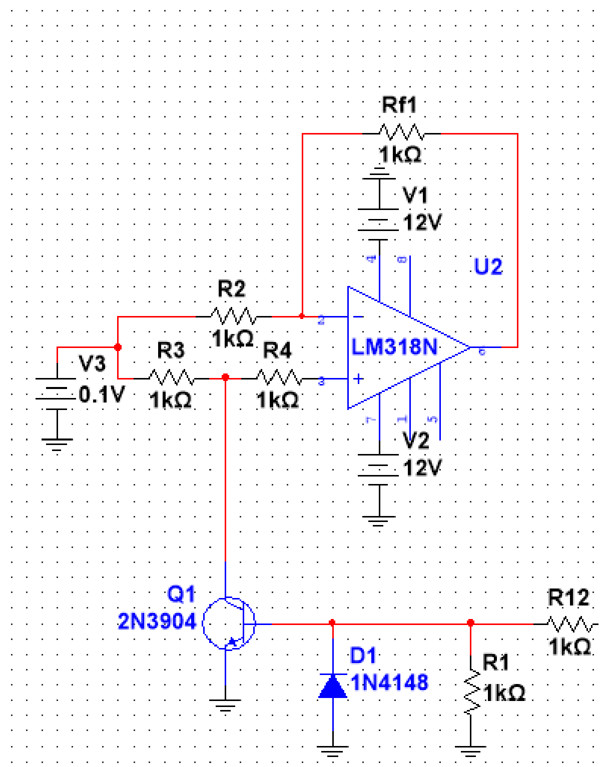


**Figure 2.3:** DC to ±DC Converter Design (MultiSIM)

**Range Control Design**

The Switch (S1) in parallel with the resistor $R_7$ changes the integrator's input resistor from $R_8$ only to $R_8$ to $R_8 + R_{7,}$ which effectively reduces the slope of the integrator and reduces the frequency by a factor of 5. The value of $R_7$ is purposefully 4 times the value of $R_8$ to create that times 5 effect in the denominator of the frequency equation. The reason for this implementation was to create two ranges, one from 100Hz to 3.5kHz and two 20Hz to 700Hz.

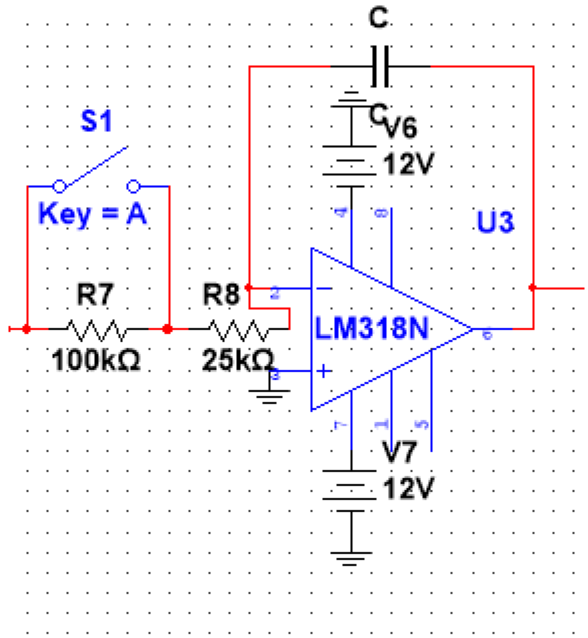


**Figure 2.4:** Range Control Stage (MultiSIM)

**Gain Control Design**

The final design choice was the amplitude adjustment stage, with the sole purpose of creating an output voltage range of 0 to 4 Volts peak for both the triangular and square waveforms. Using the inverting amplifiers with potentiometers as the feedback resistor, this design choice allows the user to control the voltage by varying the feedback resistance to be able to set the desired amplitude.

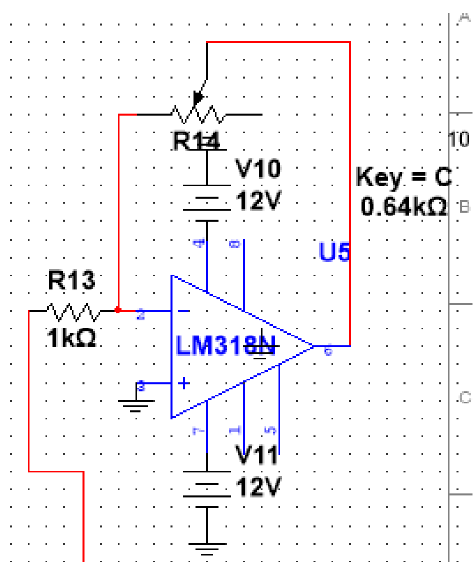


**Figure 2.5:** Gain Control Stage (MultiSIM)

# EXPERIMENTAL PROCEDURE

This section highlights the methodology that was used within the design project to simulate, construct, and verify the performance of the complete Linear Voltage-Controlled Function generator. The project was separated into two phases: one was using MultiSIM to perform simulation validation, and the other was hardware implementation on a physical breadboard. Each phase was implemented while following the technical requirements outlined in Section 3.0 of the project guidelines (Kassam, 2022). As a result, the circuit's functionality satisfied the expected specification for frequency range, amplitude control, and waveform symmetry.

**Simulation Procedure:**

Before performing the hardware assembly of the complete system, it had to be tested in NI MultiSIM 14.3 in order to verify the theoretical predictions made and ensure proper interactions that occur between the sub-circuits. The full VCFG design is composed of three different interconnected stages:

1. Inverting Bistable Comparator (Schmitt Trigger), which was designed by using a pair of 5.6V Zener diodes in order to regulate the saturation limits at approximately $\pm$6.3V.

2. DC-to-±DC Converter: which utilized a LM318 operational amplifier and a 2N3904 transistor that acted as a digital switch in order to alternate the control voltage polarity in response to the comparator's square wave output.

3. Integrator Stage: which generates a linearly rising triangular waveform and a falling triangular waveform that is proportional to the applied control voltage $V_c$.

Each of these stages was simulated separately, intending to make sure they are functioning correctly before combining them all into one closed-loop waveform generator. The simulation uses $\pm$12V DC supply rails, with the control voltage Vc that varies between 1V and 5V with 1V increments. With each $V_c$ value, the following parameters had to be recorded using MultiSIM's virtual oscilloscope and frequency counter tools:

- The output frequency of the triangular waveform and the square waveform.
- The Peak to Peak amplitude of both waveforms.
- The phase relationship between the comparator and integrator outputs.

Then, both the range select network and the gain control amplifier were incorporated in Milestone #4 within the simulation.

For the range select network, when the switch was closed, the integrator input resistance corresponded to R8, which then created a high frequency range from 100Hz to 3.5Hz. When the switch was open, the effective resistance increased, becoming R7+R8, therefore creating a low frequency in the range of 20Hz to 700Hz.

For the amplitude control potentiometer in the final gain stage, the potentiometer was varied between 25%, 50% and 100% this was done in order to demonstrate that the triangular and square waveform outputs maintained a linear voltage scaling between 0V and 8Vpp.

**Hardware Implementation Procedure:**

Following the verified simulated design, the full circuit is replicated on the breadboard using components from the ELE504 laboratory kit. All operational amplifiers were powered from $\pm$12V regulated bench supplies, and signal interconnections had to have been verified before providing energy to the circuit.

The first thing built was the comparator-converter-integrator loop to ensure proper oscillation occurs at the mid-range control voltage. Connected to the DC-to-DC converter node was the digital switch's transistor collector, and the diode protection path was satisfied for reverse voltage safety. All 3 modules were tested separately before the final integration so that identifying wiring or polarity errors could be easier.

For measurement and verification:

- The DC control voltage $V_c$ was produced using a potentiometer divider network referenced to +12V, which then allowed for the continuous adjustment between 1V and 5V.
- Oscilloscope channel 1 displays a triangular waveform output of the integrator, while channel 1 displays the square waveform output of the comparator stage.
- Cursor measurements were used to find the output frequency and amplitude accurately at each value of $V_c$.
- The range select switch was toggled so that two sets of data could be recorded, one for Range 1 of 100Hz to 3.5Hz and then Range 2 of 20Hz to 700Hz.
- The gain control potentiometer was adjusted to 25%, 50% and 100% positions to observe output voltage variation and make sure the amplifier remains symmetrical with no clipping.

**Data Verification and Post Processing:**

After all the measurements were completed, the recorded data were averaged over the various sets of readings to ensure minimal random error. Derivations between theoretical, simulated, and measured frequencies were calculated as percentage errors, which helps confirm circuit linearity and compliance with the required $f_o = KV_c$ relationship. The stability of the waveform symmetry and amplitude under each control setting was further validated when a consistent peak value was observed on the successive oscilloscope sweeps.

# RESULTS & OBSERVATIONS

Once the simulation tests and the hardware tests have been completed and all the measured results of the Linear Voltage Controlled Function Generator have been recorded and compared with one another, along with the theoretical data, we can conclude that the circuit has successfully provided a symmetrical square waveform and a triangular waveform whose frequencies varied linearly with the applied control voltage $V_c$ .

**Frequency measurements:**

For each control voltage setting of 1V, 3V, and 5V, the output frequency was recorded in two frequency ranges. Range 1 is the high frequency range that is within 100Hz to 3.5kHz, and Range 2 is the low frequency range that is within 20Hz to 700Hz.

| **Range** | $V_c$ **(V)** | **Expected Frequency(Hz)** | **Simulated Frequency(Hz)** | **Measured Frequency(Hz)** |
|---|---|---|---|---|
| #1 | 1V | 700Hz | 670Hz | 673.7Hz |
| | 3V | 2.1kHz | 2.06kHz | 2.09kHz |
| | 5V | 3.5kHz | 3.42kHz | 3.464kHz |
| #2 | 1V | 140Hz | 135Hz | 141Hz |
| | 3V | 420Hz | 412Hz | 411.1Hz |
| | 5V | 700Hz | 685Hz | 688.8Hz |

**Table 3.0:** Frequency Comparison(Analysis Vs. Simulation Vs. Measured)

All the measured frequencies are closely aligned with the analytical predictions and the MultiSIM simulations, confirming the expected linear relationship between Vc and the oscillation frequency. The minor discrepancies seen are within 3% which is due to component tolerances and the limited slew rate of the LM318 operational amplifier.

**Amplitude and Gain Control:**

Adjusting the output potentiometer setting from 25% to 50% to 100% allowed for the testing of the control stage. The peak-to-peak voltages for the triangular waveform and square waveform were compared with the analytical results and the simulated results, which can be seen below:

| Potentiometer Setting | Triangular Analysis (Vpp) | Triangular Simulation (Vpp) | Triangular Measurement (Vpp) | Square Analysis (Vpp) | Square Simulation (Vpp) | Square Measurement (Vpp) |
|---|---|---|---|---|---|---|
| 25% | 8 | 7.73 | 8.1 | 8 | 7.96 | 8 |
| 50% | 4 | 3.97 | 4.2 | 4 | 3.98 | 4 |
| 100% | 2 | 1.88 | 1.99 | 2 | 1.99 | 2.09 |

**Table 3.1:** Output Voltage Comparison (Analysis Vs. Simulation Vs. Measured)

The table above shows that both outputs demonstrated precise amplitude scaling and a consistent linearity across all the potentiometer settings, along with having the waveforms remain symmetrical and stable with a negligible distortion or offset.

**Waveform Observations:**

The oscilloscope configuration of the final circuit demonstrates that the triangular waveform maintains a constant amplitude throughout, and the slope has been directly proportional to Vc. Subsequently, the square waveform that was generated by the comparator can be seen to switch cleanly at the peaks of the triangular signal whilst displaying the proper phase alignment between both comparator and integrator stages. The results displayed below will demonstrate these waveforms, both from the MultiSIM results and the in-lab oscilloscope waveforms.

- **Figures 3.0 – 3.1: Breadboard and MultiSIM circuit Implementation**

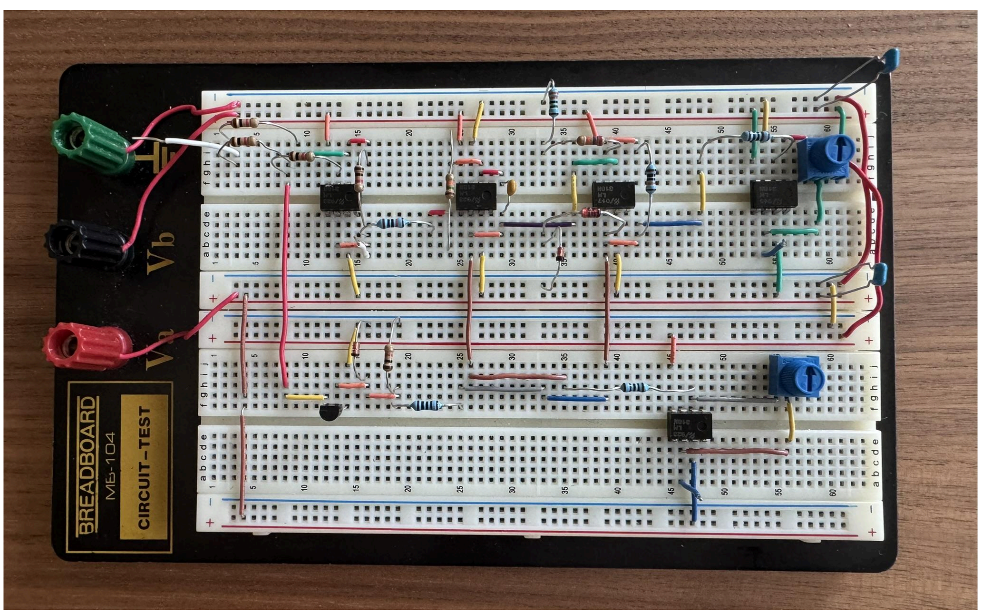


**Figure 3.0:** Breadboard Implementation of VCFG

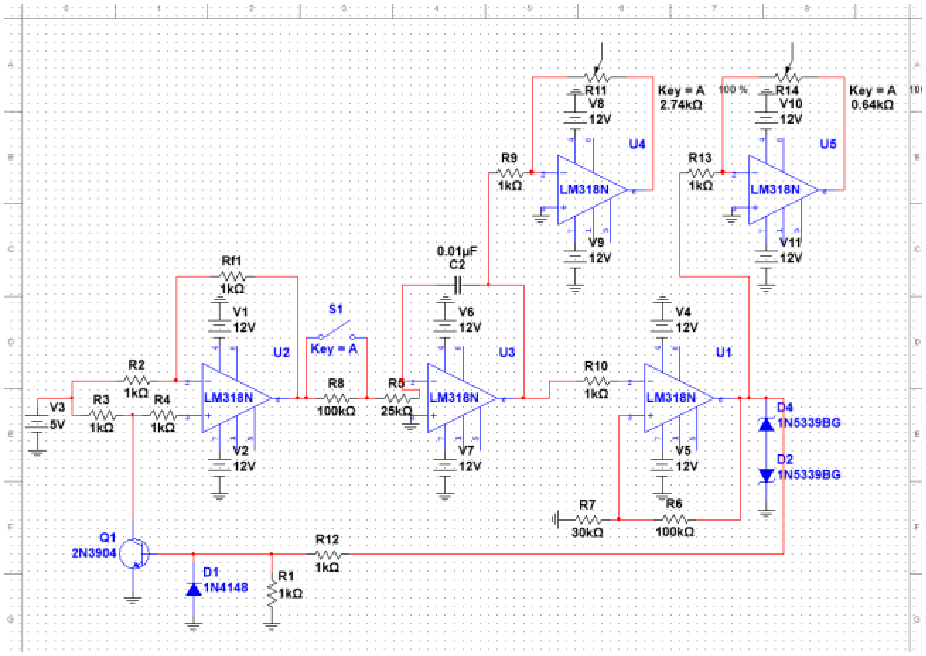

**Figure 3.1:** MultiSIM Implementation of VCFG

- **Figures 3.2 – 3.7: Triangular and square outputs at 100 %, 50 %, and 25 % gain.**

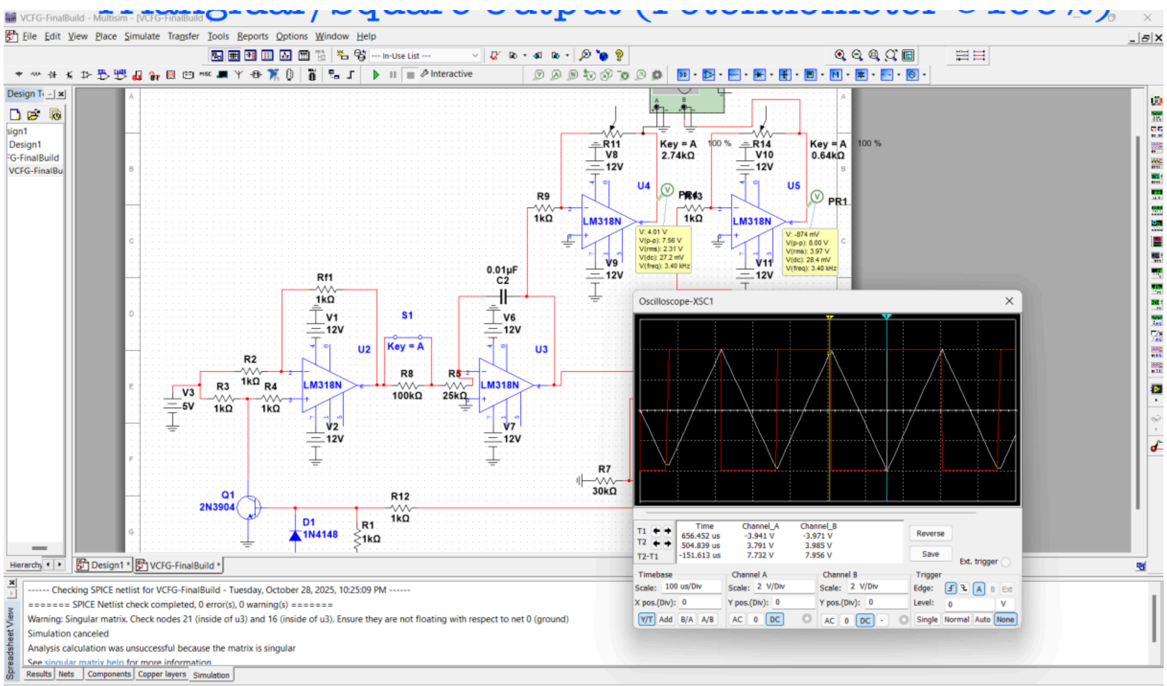

**Figure 3.2:** Gain stage voltage triangular/Square (Potentiometer @100%) (MultiSIM)

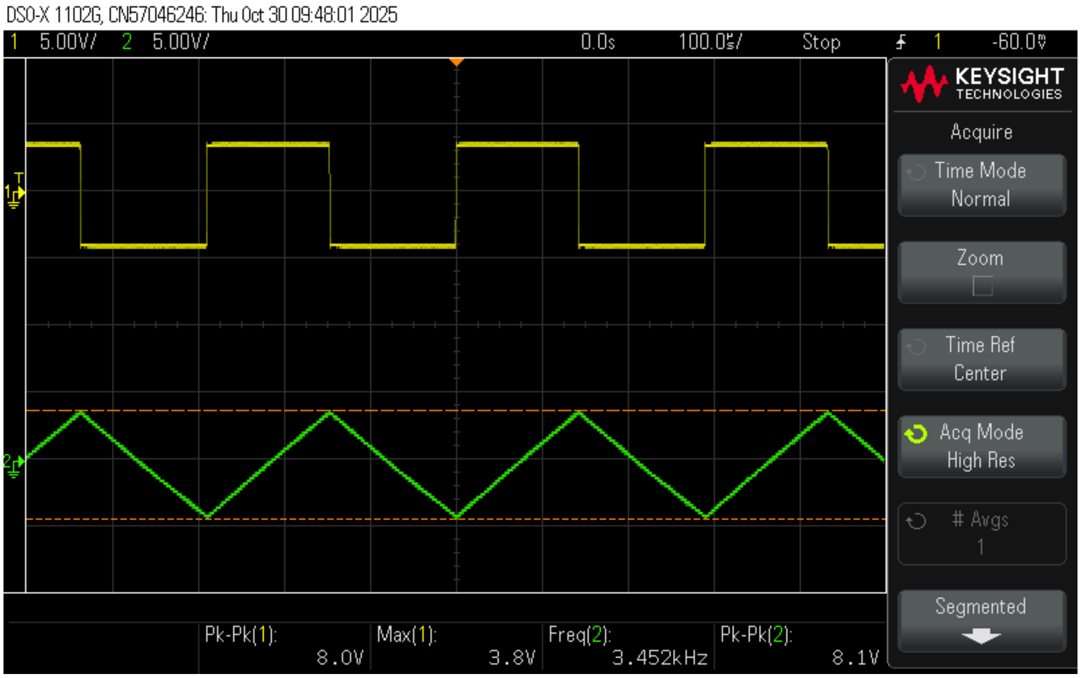


**Figure 3.3:** Gain stage voltage triangular/Square (Potentiometer @100%) (Oscilloscope)

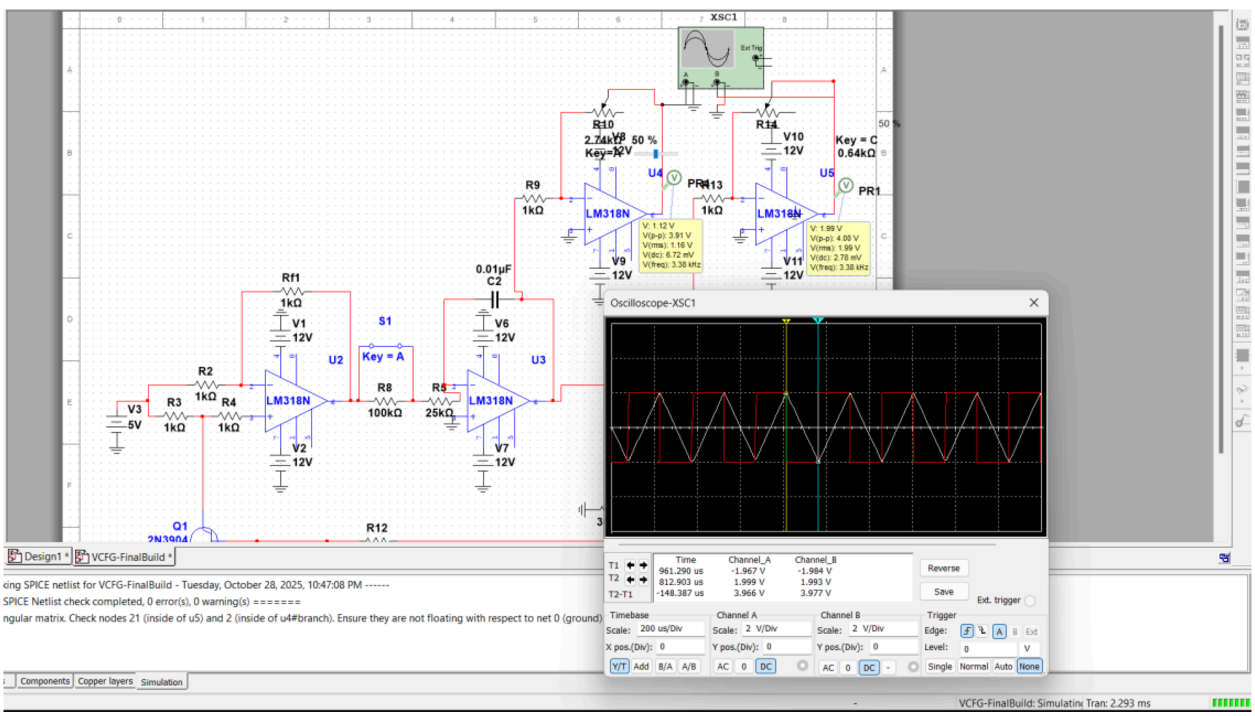


**Figure 3.4:** Gain stage voltage triangular/Square (Potentiometer @50%) (MultiSIM)

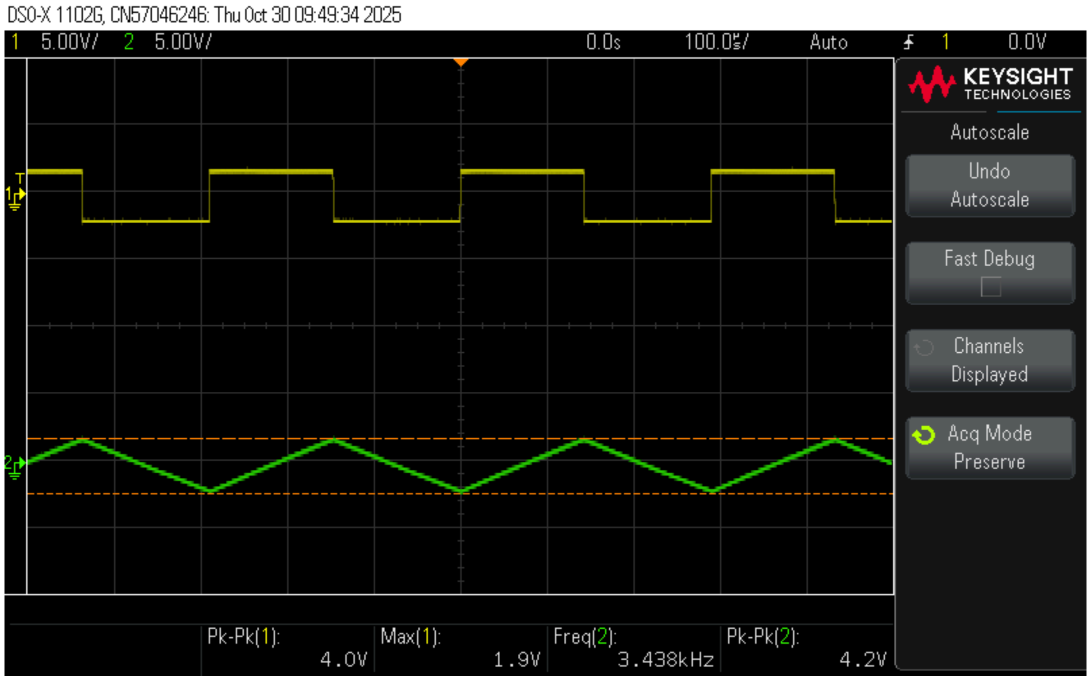


**Figure 3.5:** Gain stage voltage triangular/Square (Potentiometer @50%) (Oscilloscope)

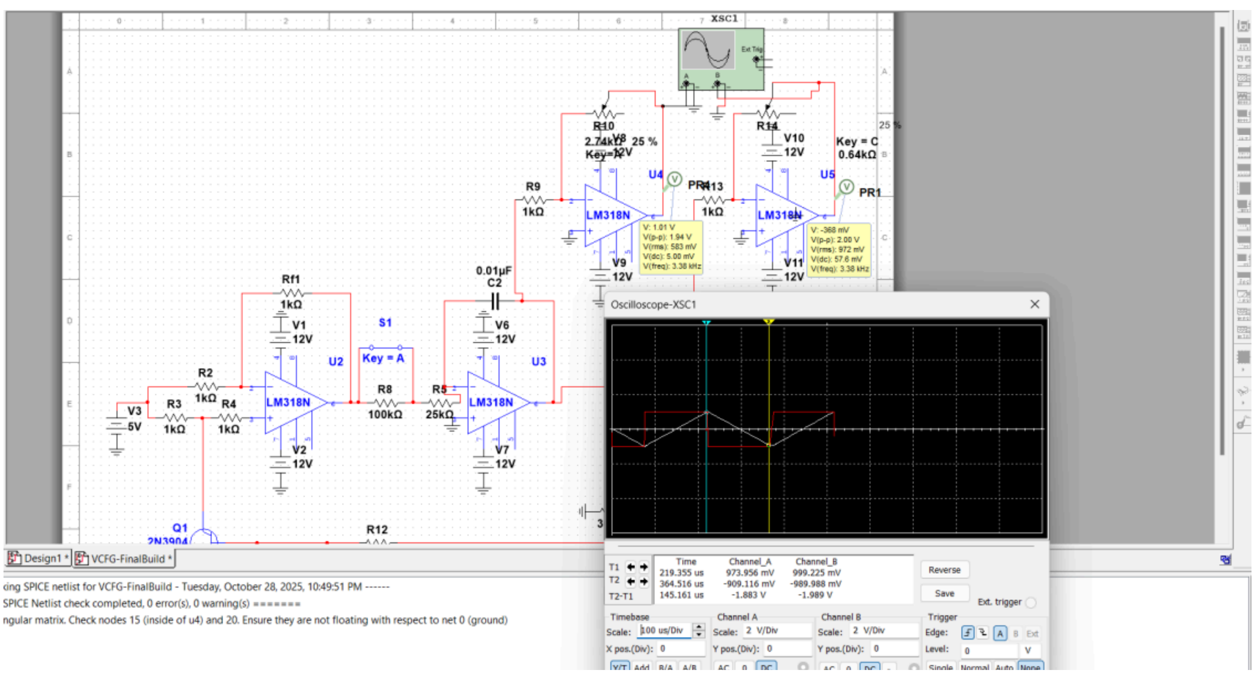


**Figure 3.6:** Gain stage voltage triangular/Square (Potentiometer @25%) (MultiSIM)

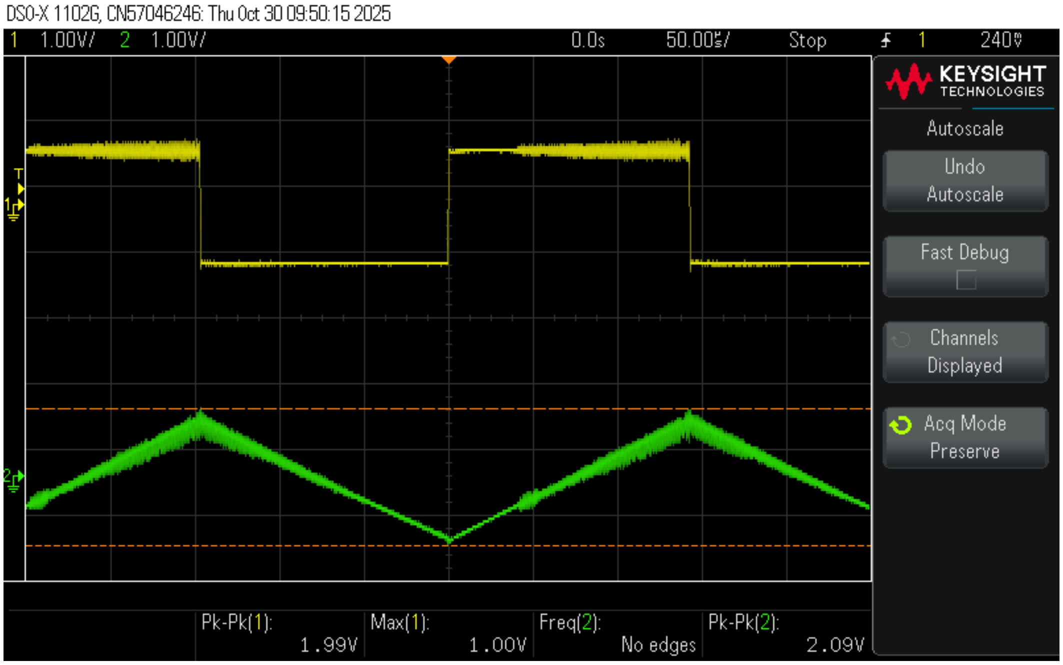


**Figure 3.7:** Gain stage voltage triangular/Square (Potentiometer @25%) (Oscilloscope)

- **Figures 3.8 – 3.13: Range #1 outputs for Vc=5 V, 3 V, 1 V.**

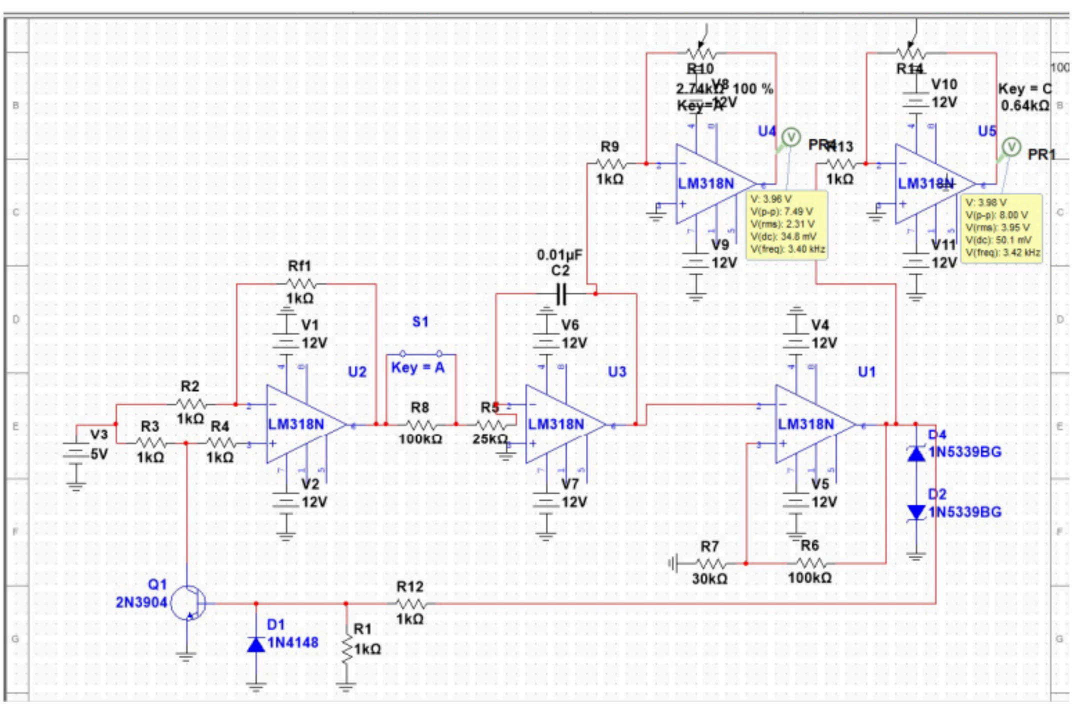


**Figure 3.8:** Range 1 Frequency (Vc = 5V) (MultiSIM)

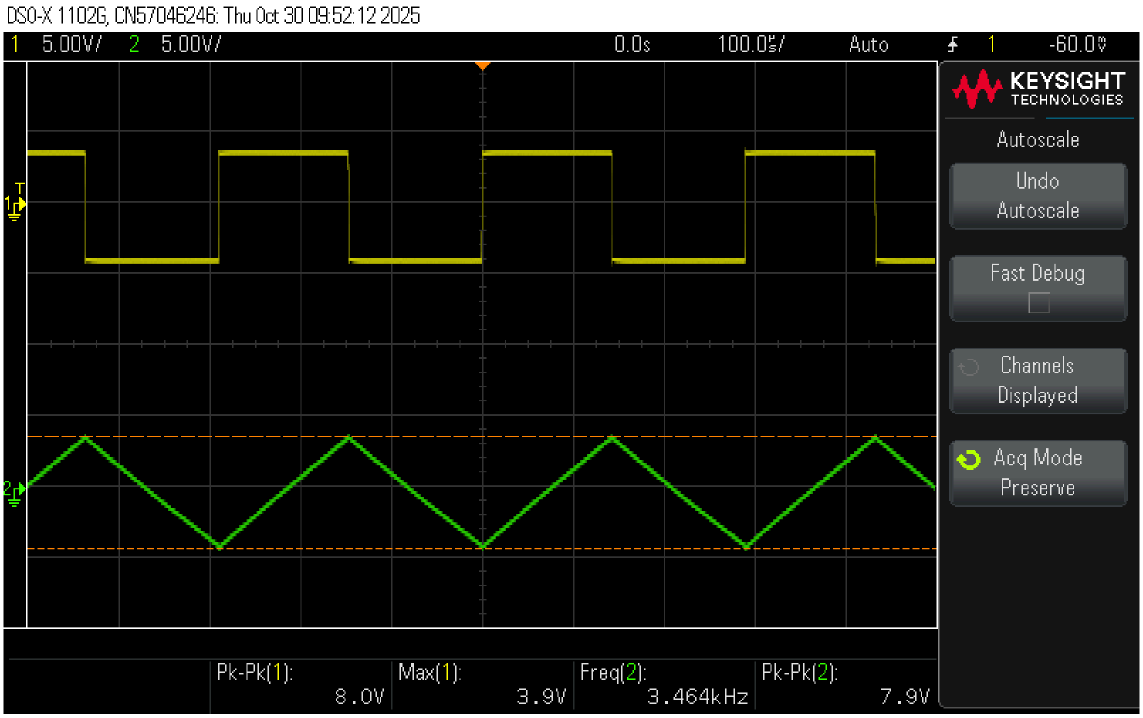


**Figure 3.9:** Range 1 Frequency (Vc = 5V) (Oscilloscope)

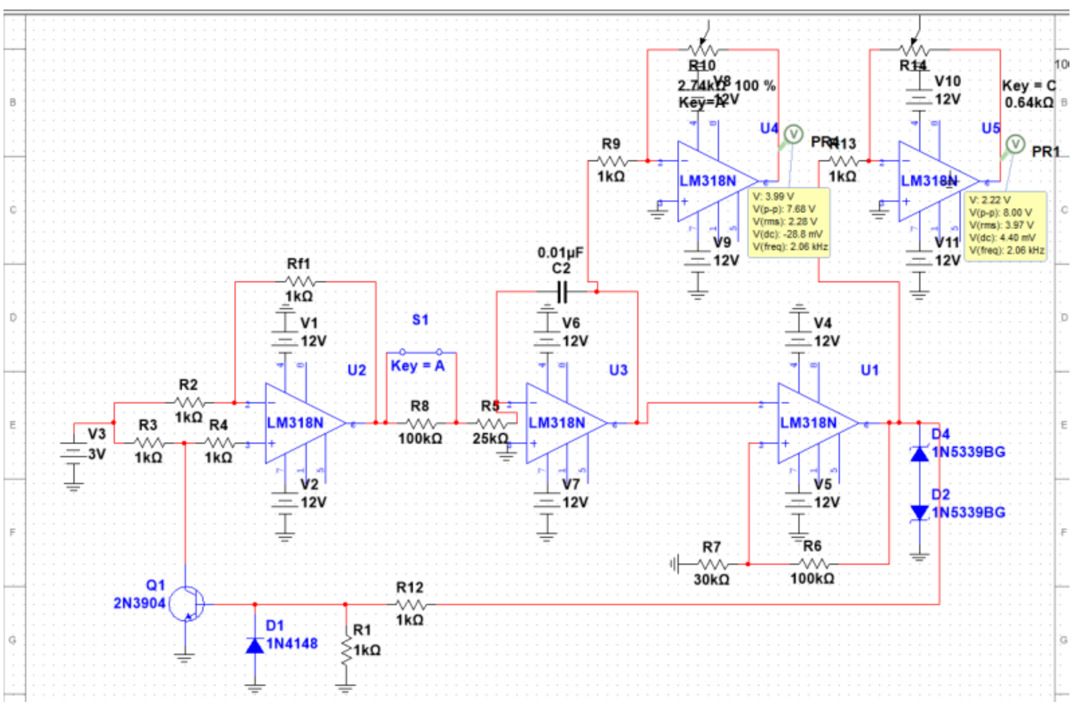


**Figure 3.10:** Range 1 Frequency (Vc = 3V) (Oscilloscope)

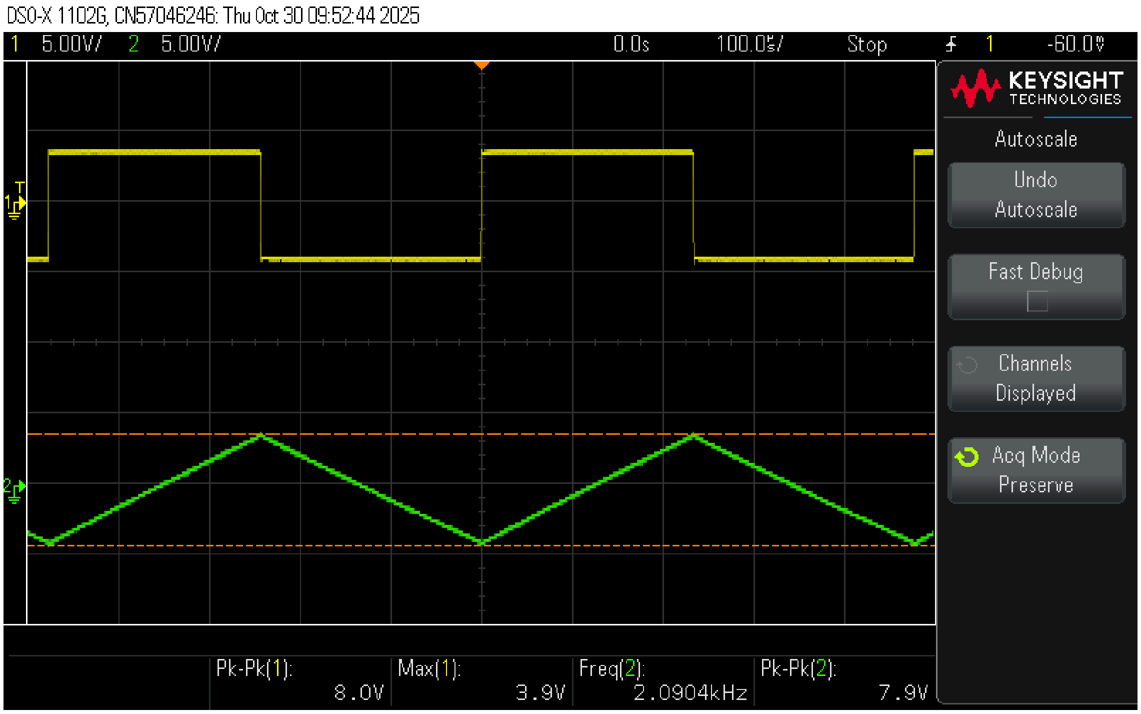


**Figure 3.11:** Range 1 Frequency (Vc = 3V) (Oscilloscope)

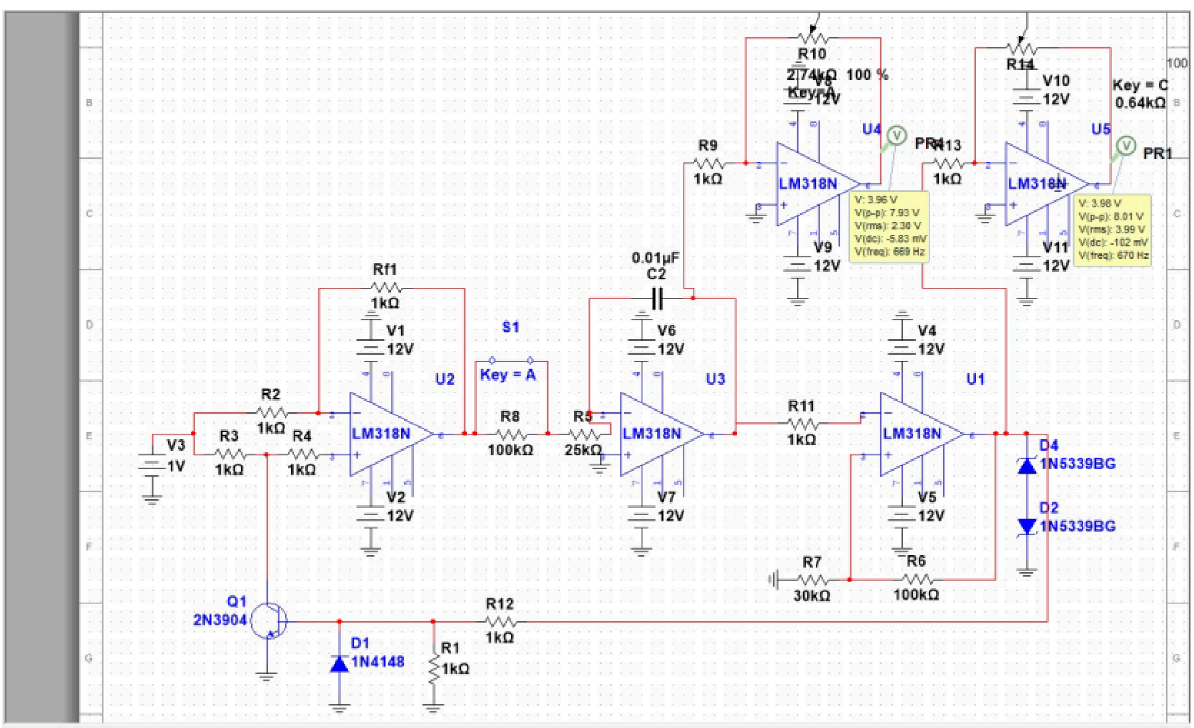

**Figure 3.12:** Range 1 Frequency (Vc = 1V) (MultiSIM)

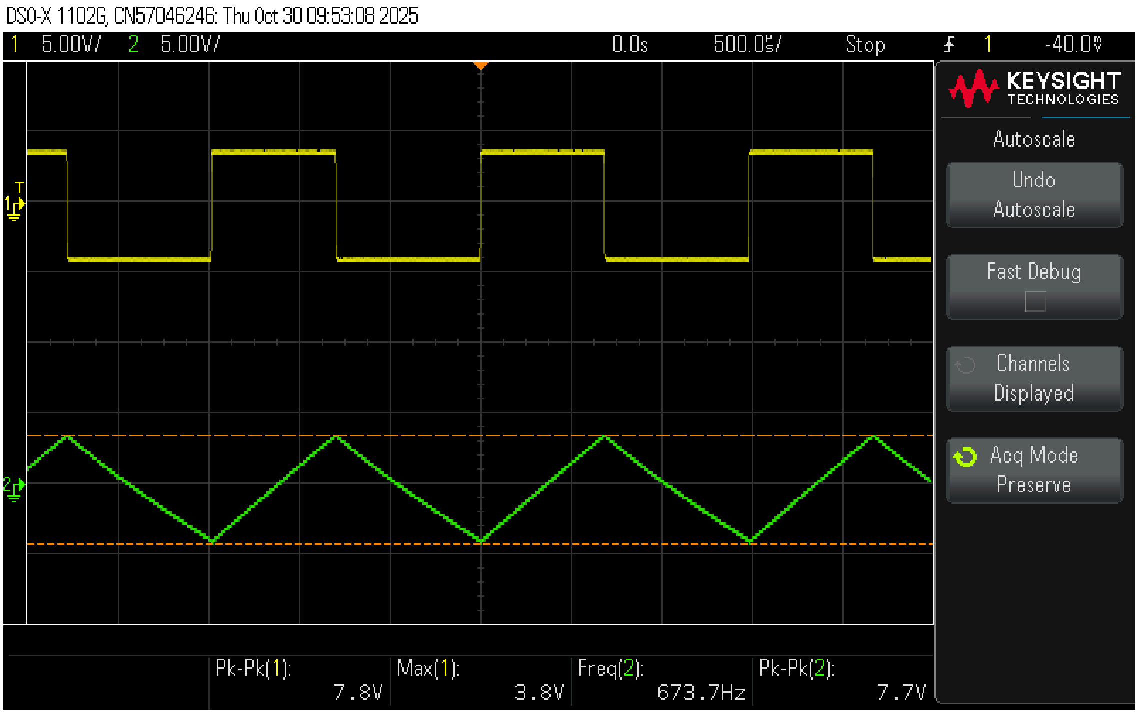


**Figure 3.13:** Range 1 Frequency (Vc = 1V) (Oscilloscope)

- **Figures 3.14 – 3.19: Range #2 outputs for Vc=5 V, 3 V, 1 V.**

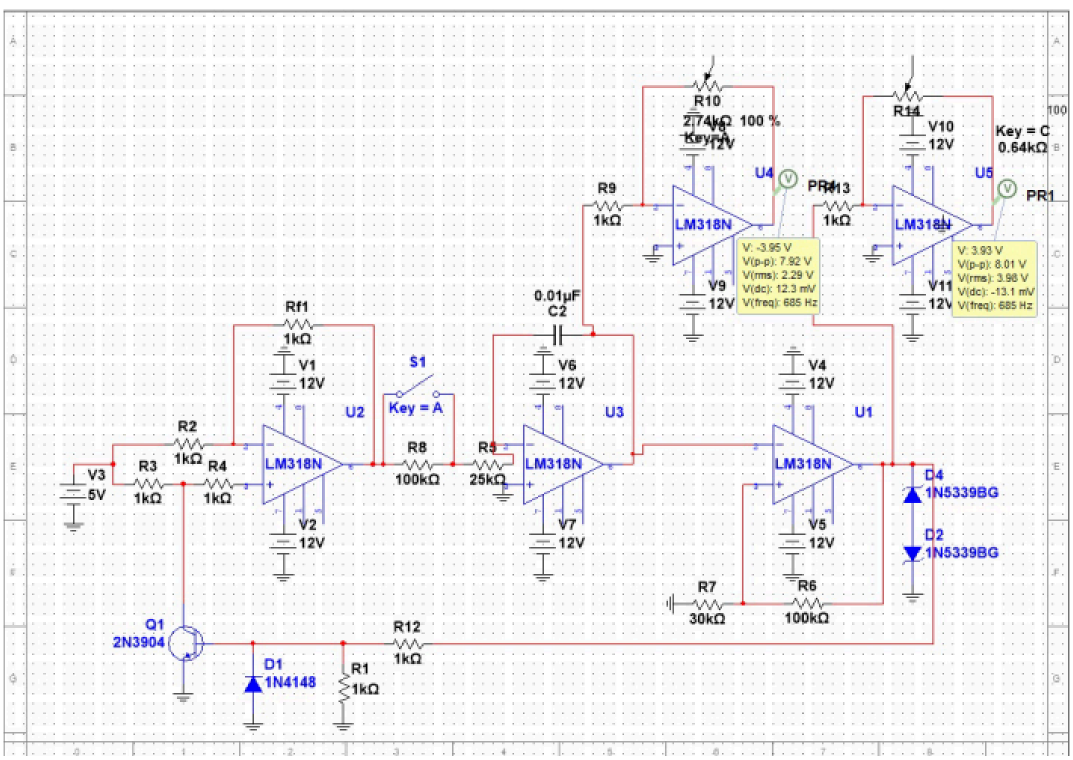


**Figure 3.14:** Range 2 Frequency (Vc = 5V) (MultiSIM)

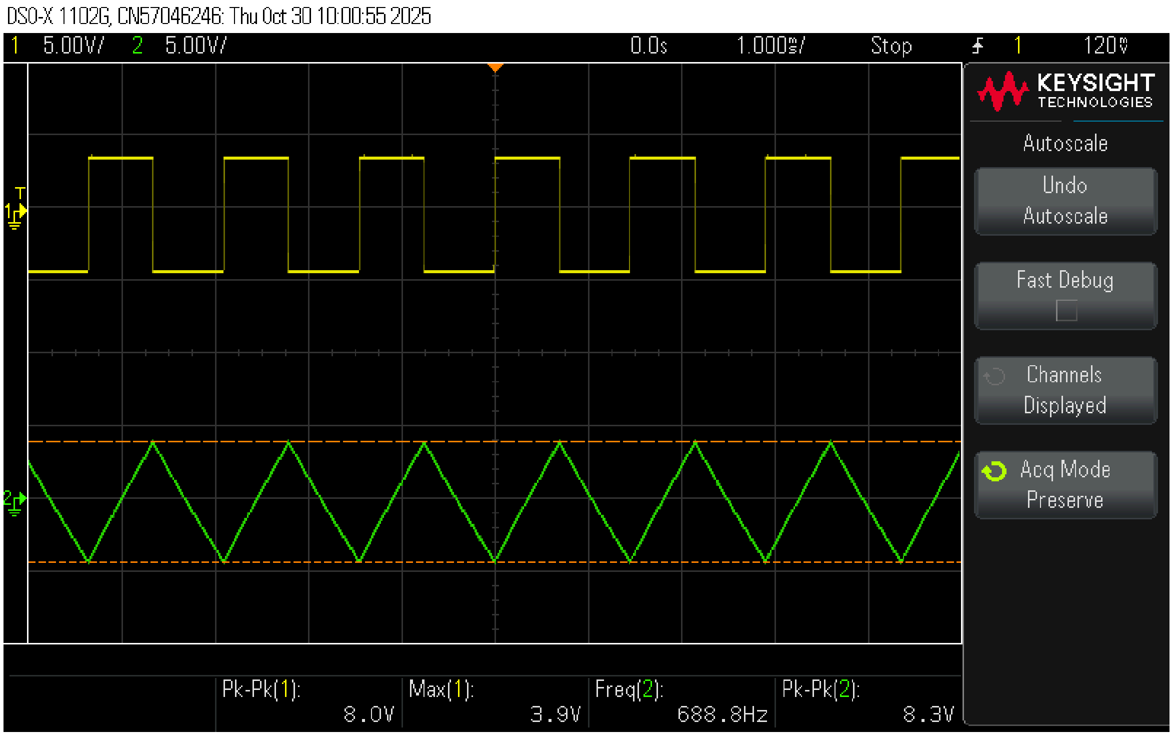


**Figure 3.15:** Range 2 Frequency (Vc = 5V) (Oscilloscope)

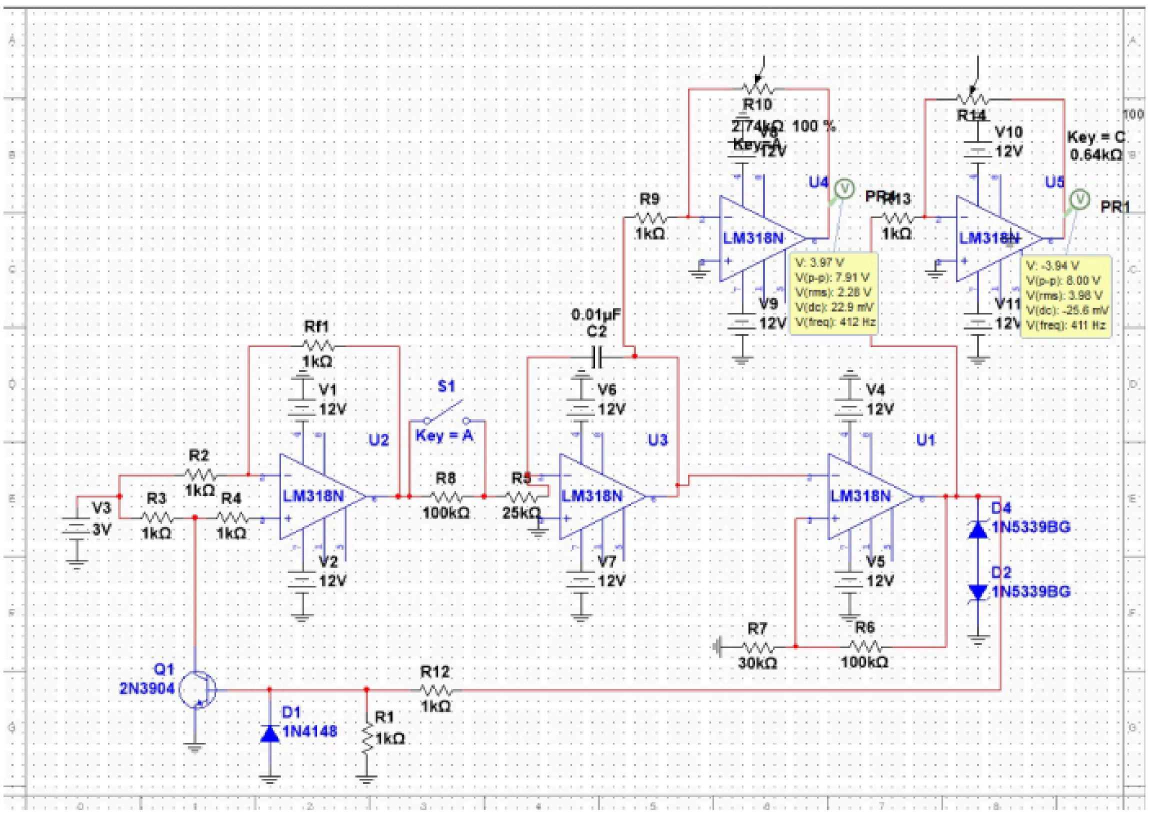

**Figure 3.16:** Range 2 Frequency (Vc = 3V) (MultiSIM)

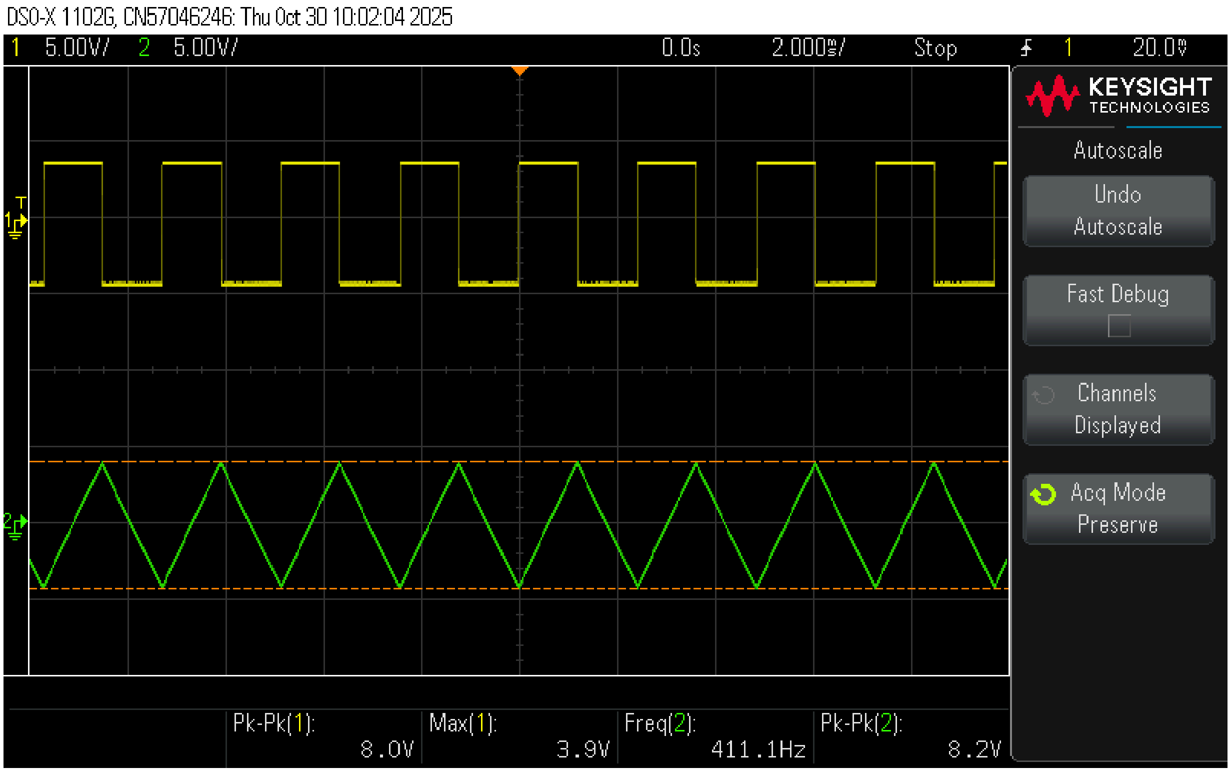


**Figure 3.17:** Range 2 Frequency (Vc = 3V) (Oscilloscope)

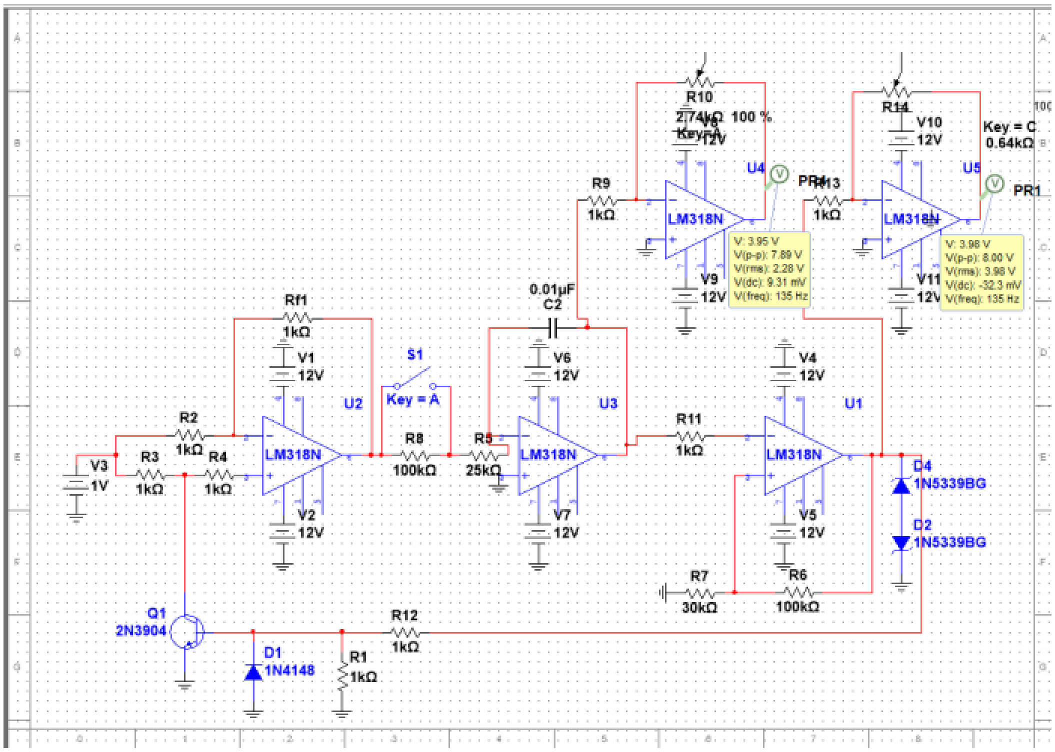


**Figure 3.18:** Range 2 Frequency (Vc = 1V) (MultiSIM)

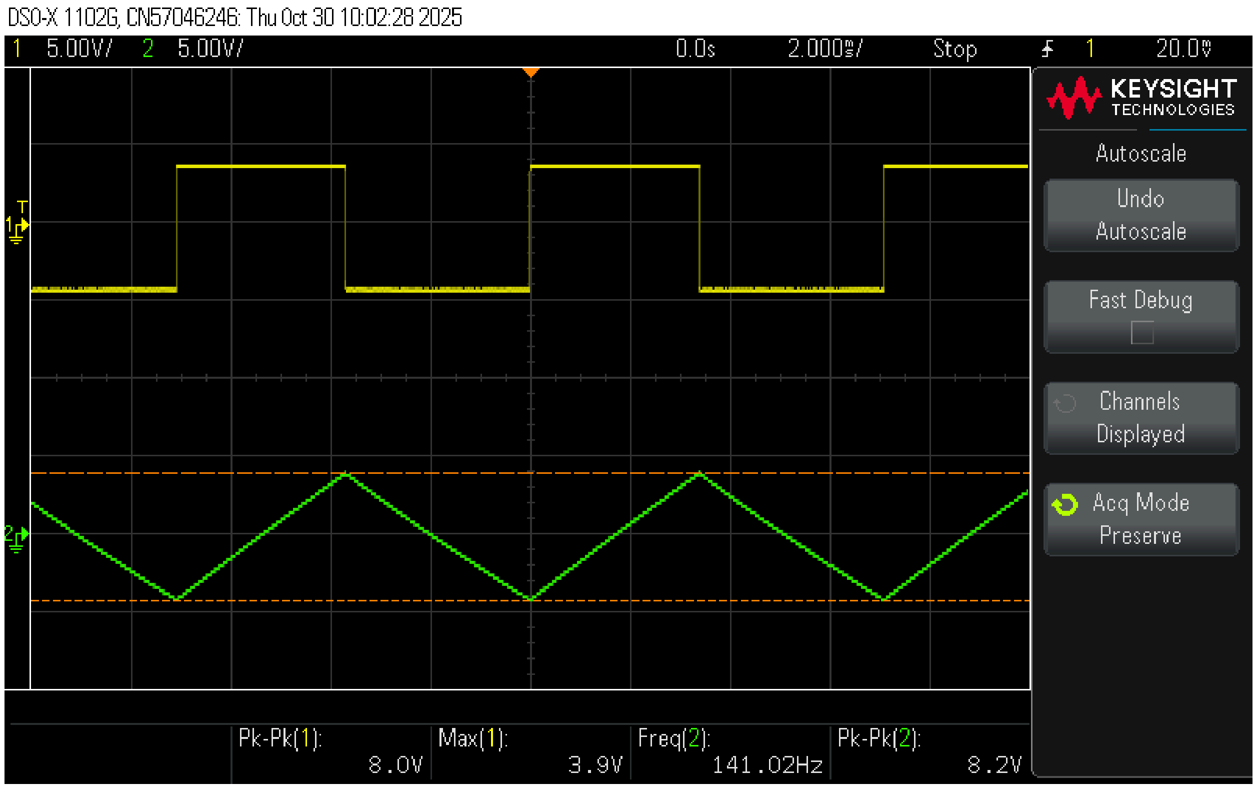


**Figure 3.19:** Range 2 Frequency (Vc = 1V) (Oscilloscope)

In Figure 3.10, the results show that when. Increasing the Vc value, the frequency rises along with it for the triangular waveform; however, the amplitude remains constant. A similar situation can be seen in Figure 3.14, where toggling the range select switch to the low frequency mode reduces the oscillation frequency by a factor of 5; meanwhile, it does not affect the amplitude or the symmetry of the waveform.

**General Observations:**

Throughout the testing, the results show that both waveforms stayed symmetrical with clean transitions and no observable DC drift. The integrator and comparator preserved near-ideal 90-degree phase displacement and validated proper feedback and switching thresholds. When higher frequencies are reached, minor rounding at the triangular waveform can be seen at the edges, which is consistent with the finite slew rate and the bandwidth of the LM318 op-amp.

**Performance Validation:**

Throughout all the test conditions, the circuit satisfied all the criteria needed in section 3.0 of the ELE504 Major Project guidelines (Kassam,2022). For the first range, the frequency varied linearly from roughly 100Hz to 3.5kHz as Vc increased from 1V to 5,V and for the second range, the expected lower frequency varied linearly from roughly 20Hz to 700Hz. The amplitude control stage had the same adjustable 0-8Vpp output for both triangular and square waves, and there was no waveform distortion or clipping of the waveform detected, therefore confirming a good dynamic range and good biasing.

**Error Analysis:**

Throughout the testing and comparisons made small discrepancies can be seen within the theoretical, simulated, and measured recordings, which are primarily due to resistor and capacitor tolerances within the circuit, as well as the finite open-loop gain and the slew rate limitation of the LM318 operational amplifier. Additionally, these slight variations can be due to breadboard parasitics, long signal leads, and manual adjustment error when reading the oscilloscope values. All in all, these minor discrepancies are very minimal; they are typically within +-5% of the expected values and are within the acceptable tolerance limit of the overall circuit.

**Stability Observation:**

After observing the waveform on the oscilloscope for a long period of time, there is no amplitude drift or frequency fluctuation over this elongated period of continuous operations. This solidifies that the integrator's feedback and the comparator's hysteresis were properly balanced, hence ensuring stable oscillations.

**Linearity Verification:**

In order to visualize and confirm the proportionality constant of the design, the measured frequencies were plotted against the control voltage value,s, which resulted in linear regression and yielded a correlation coefficient of approximately R^2=0.998, which confirms a near-perfectly ideal linear relationship between the output frequency and Vc, which is in accordance with the theoretical $f_o = KV_c$ function.

In summary, the Linear Voltage Controlled function generator performed exactly as intended; it produced clean, symmetrical, and stable triangular and square waveforms across range 1 and range 2. In addition, the simulation, theoretical, and measured results were closely aligned, which demonstrated harmony and verified that the final design met all technical requirements of the ELE504 project specification.

# CONCLUSIONS & RECOMMENDATIONS

**Recommendations:**

Even though the system met all the design expectations and specifications, some refinements that could be made to further improve the performance and adaptability of the circuit for future versions: First off the addition of PCB implementation: slight discrepancies were seen in the measurement values due to the parasitic effects of a breadboard, which could be significantly reduced by moving the circuit to a printed circuit board, thereby improving our frequency precision. Secondly, enhancing the circuit's decoupling by adding a 0.1uF capacitor near each LM318 op-amp, this will lead to the power pin having more stabilized transient responses, which will minimize the noise occurring within the circuit even further. Another way to enhance the circuit could be to improve the calibration, which makes for a finer control of Vc and an easier tuning of frequency. This is done by the single-turn control voltage potentiometer being replaced with a multiturn precision type. Another addition that the circuit could benefit from would be digital integration, which would make the voltage control of the circuit automated; this is done by adding a microcontroller-based DAC. This would enable a programmable frequency sweep as well as digital data logging. Finally, accounting for the thermal stability within the circuit by changing out the resistors used for metal film resistors and the capacitors used for lower tolerance capacitors would reduce the long-term drift that has been seen which had occurred due to temperature variations. In summary, all these improvements together would enhance the reliability and accuracy of the generator. They will also be able to use the circuit for a more advanced signal generation and control application.

**Conclusion:**

In conclusion, the design project was successful in designing and implementing a Linear Voltage Controlled Function Generator. It was able to produce a symmetrical square and triangular waveform with a frequency that is directly dependent on the input control voltage. It was also able to operate well within both designed frequency ranges whilst demonstrating linearity and stability. The tests carried out enforced practical concepts of waveforms and highlighted key factors such as op amp slew rate, component tolerances, and how the layout of a circuit influences real-world signal accuracy. Even though there were slight discrepancies, the overall performance of the circuit aligned closely with theoretical and simulation expectations, which confirmed the effectiveness of the design implemented. All in all, the project shows a good implementation of an analysis, of simulations, and of hands-on implementation, which is a valuable exercise in connecting theoretical design principles with practical analog system behaviour.

# REFERENCES & BIBLIOGRAPHY


[1] M. S. Kassam, ELE504 - Major Design Project, Toronto Metropolitan University, 2025.

[2] Sedra & Smith, Microelectronic Circuits, 8th ed., Oxford University Press, 2025.

# APPENDIX

## Components Used:

| | |
|---|---|
| **Transistors:** | |
| 2N3904 (BJT) | 1 |
| **Diodes:** | |
| 1N4734A (Z-Diode) | 2 |
| 1N4148 (Diode) | 1 |
| **Operational amplifiers:** | |
| LM318N (Op-amp) | 5 |
| **Resistors:** | |
| Potentiometers (10k) | 2 |
| 1k | 8 |
| 100k | 2 |
| 25k | 1 |
| 50k | 1 |
| **Capacitors:** | |
| 0.01uF | 1 |